

The 20-Day MVP — Complete Scope Definition (Knowledge Base 08)

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SCOPE LOCKED — final agreed position

Timeline	20 working days (was 14)
Team	2 developers on web · 1 developer on mobile
Features	35 of 51 agreed IN
Integrations	6 — Xero · Twilio · Google Maps · MS365 · MS365 Outlook · Mistral AI OCR
Authentication	Email OTP · SMS OTP
Driver app	★ Flutter native AND React PWA — both complete. Resolves background GPS
Stack	MongoDB Atlas (Sydney) · Express · React + Vite · Flutter · BullMQ + Redis · Render — see §6A

What changed from the 14-day scope:

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IN: F22 (driver performance) · F43 (vehicle maintenance + rego) · F46 (preferred pickup times, restored) · F53 (driver training records) · F65 (2-way reschedule)

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IN: I6 MS365 Outlook + I8 Mistral OCR → the automated PO-from-email pipeline moves from v1.1 into scope. This was the client's stated core pain

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OUT: F27 (post-job follow-up emails) · F41 (driver job feedback) · F32 (customer document upload)

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Restored: M9.6 financial summary — the 20-day budget absorbs it

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Risk 7 resolved by decision: native app for background GPS, PWA alongside it

What this document is:

the single end-to-end definition of what we will build, ship and put into production in 20 days — organised by **module** (the big things), with every **feature**, **integration**, **authentication method** and **user workflow** explicitly marked IN or OUT, described in plain language, and illustrated with a concrete example.

What it is for:

it is simultaneously (a) the scoping decision, (b) the build brief for the dev team, and (c) the basis of the conversation with Matthew Browne about what he is and is not getting on day 20.

The governing constraint, stated plainly:

this is **not** a greenfield build. It is the **replacement of a working TMS (TransVirtual) that has processed 5,079 jobs and currently invoices ~\$55–65k per month**. On day 21 the business must still bill correctly. Every scope decision below is made against that constraint, not against the wish list.

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1. The honest headline

35 of 51 features, in 20 working days, with 3 developers. That is now the agreed scope, and it is materially more ambitious than the original 14-day cut.

The full original ask was 51 features × 7 personas × 8 integrations × 109 listed workflows, plus a WYSIWYG report designer, plus migration of three years of data. What has been agreed is a deliberate, negotiated subset — and the extra six days bought three things that were previously deferred: the **AI PO-ingestion pipeline**, a **native driver app**, and the **fleet/driver-compliance features** (F22, F43, F53).

What ships in 20 days is a complete, coherent product that:

1. **Replaces the six screens PlastaGo actually live in every day.** We know exactly which six, because they pinned them as favourites in TransVirtual: create a job, find a job, approve driver charges, action futile pickups, invoices, where are my drivers (06– 53).
2. **Delivers the two things TransVirtual cannot do** — a real customer portal, and automated recycling certificates.
3. **Bills correctly from day one**, validated against their own historical invoices before we switch anything over.

Eleven modules. 35 of 51 features. 78 of 109 workflows. Everything else is explicitly deferred with a reason and a version number.

⚠️ **The honest read on capacity.** 20 days × 3 devs ≈ **60 developer-days**. The added scope (AI PO pipeline, native app, F22/F43/F53) is roughly **10–13 developer-days**, against **+18 developer-days** of extra capacity. It fits — but not loosely, and two items carry external dependencies we do not control: **Apple Developer enrolment** (Risk 8) and **OCR extraction accuracy** (Risk 9).

2. Terminology — clearing up three confusions in the brief

The feature list contains some wording that will cause the wrong thing to be built if taken literally. Resolving these **reduces** scope while **increasing** what the client gets.

2.1 Two different apps, two different answers

The decision is now split by audience — and it is worth stating clearly because the brief used "mobile app" for both.

Surface	Build	Why
Customer portal (F21)	PWA only	Site supervisors will not install anything. Matt: <i>"I don't think anyone's going to install a PlastaGo app onto their phone — it's like a website that's phone compatible"</i>
Driver app	★ Native AND PWA	Background GPS. Not reliably available to a PWA and effectively unavailable on iOS — and Troy is on an iPhone. The native build solves it; the PWA is the fallback and the fast-iteration path

F21 specifically still means PWA. The customer-facing "mobile app" is the portal (M5), mobile-responsive and installable to the home screen. **No native customer app.**

There is no need for a native iOS/Android app **for customers**, and building one would be a mistake (app-store review alone can consume a meaningful slice of the sprint).

What we build instead: the **Customer Portal** (M5) as a **Progressive Web App** — a mobile-responsive web application that a site supervisor opens in their phone browser and can "Add to Home Screen" so it behaves like an app (own icon, full screen, no browser chrome, works offline for viewing).

Why this is better, not a compromise:

- **Zero install friction.** A site supervisor receives an SMS with a link, taps it, logs in with a one-time code, and books a pickup. No app store, no password, no IT involvement.
- **Instant updates.** No release cycle, no users stuck on old versions.
- **One codebase** for desktop office users and phone-based site supervisors.
- The client's own spec already says **"v1.0 Progressive Web App"** on the cover page — so this matches their stated intent; only the F21 wording is loose.

Therefore F21 is not a separate feature. It is absorbed into M5.

The driver app is the exception. It ships as **both**: a native shell (for background location, and for reliable camera/storage on older devices) **and** the same PWA underneath. Both are **offline-first** — drivers work in greenfield estates with poor coverage. See M4 and Risk 8 for the distribution mechanics.

2.2 "Feature" vs "Module" — how this document is organised

The 51-item feature list is flat, which makes it look like 51 separate things to build. It isn't. Most items are **capabilities within a larger surface**.

For example, these seven "features" are all just *the Customer Portal*:

F9 (customer portal) · F18 (view job history) · F21 (mobile app to submit requests) · F26 (customer dashboard) · F28 (request urgent pickup) · F32 (submit documents) · F46 (set preferred pickup times)

Building "the Customer Portal" once delivers all seven. This document therefore groups everything into **10 modules**, each with a description of what the module *is*, followed by its individual feature items with descriptions and examples.

2.3 Many "workflows" are the same screen viewed by a different role

The workflow list includes, separately:

W65 — *Office Worker · tracking job delays and reasons* W108 — *Allocator · Track job pickup delays*
W120 — *Allocator · tracking job delays and reasons* F30 / F54 — *tracking job delays / pickup delays and notifications*

These are **one capability** (delay reason capture and reporting) appearing four times. Roughly **80 of the 109 listed workflows are permutations of ~30 real screens**.

Do not treat the workflow list as a backlog. Treated literally it triples the apparent scope for zero added function. Section 10 maps every W-number to the module that delivers it.

3. Go-live strategy: staged cutover

Phase	What happens
Days 1–20	Build · migrate · run the pricing regression harness continuously
Day 20 — GO LIVE	The new system becomes <i>the</i> system for intake, allocation, driver execution, customer portal, notifications and invoicing → Xero
Days 21–34	TransVirtual retained read-only as safety net and historical archive. Every invoice spot-checked in week 1
~Day 34	TransVirtual cancelled

Why not kill TransVirtual on day 20: it costs a few hundred dollars for a fortnight, and it removes the single catastrophic risk — discovering an undocumented pricing rule *after* cutover with no fallback and no way to bill. This is professional practice, not hedging, and an operations director will respect it being said out loud.

4. The three scope calls worth arguing for

These are the three places where we should **push back on the brief**. Each one saves days, and each one gives the client a better outcome than what they asked for. They should be agreed **in writing before build starts**.

Scope Call 1 — Split the invoice template requirement in two

What Matt asked for

"They also want to be able to create and maintain custom Invoicing Templates... sometimes even on a per-job basis." "Below is a screenshot of what the report designer looks like. **I don't need an exact replica of this, but being able to change and edit all the same things is important to us.**"

The tool in that screenshot is **Stimulsoft Reports** — a full commercial banded report designer with data sources, business objects, variables, functions, expression bands and drag-and-drop layout. Rebuilding it is a product in its own right, not a feature.

⚠️ REVISED after Call 2 — Matt wants the opposite split

An earlier version of this call recommended "build resolution, defer authoring." **Call 2 shows he wants close to the reverse** (09- §21):

On resolution — he does NOT want it:

"I **don't need this level of granularity** necessarily. Once we pick an invoice, we're pretty much — most of our customers will comply."

On authoring — he does, and he built the current templates himself:

"Will I be able to **edit the layout of my invoices** in this system?" "This is obviously something I've set up before all myself just to suit our needs."

He also uses the same designer for **operational reports**, not only invoices.

The revised split

Capability	Verdict	Rationale
Template resolution — priority + data filter	⬇️ SIMPLIFY to per-customer template assignment	Matt explicitly doesn't need the granularity. This is now cheaper than originally planned — assign a template to an account, done
Branding & content control — logo, colours, company details, terms, bank details, footer, visible columns	✅ IN — 14 days	Still the bulk of the real editing need
Template authoring — layout editing	⚠️ DECISION REQUIRED — see below	Genuinely wanted. Full band designer is weeks

Three honest options for authoring, to put to Matt

1. **Constrained editor** — logo, colours, company details, terms, bank details, footer text, visible columns, free text blocks. Covers most real edits. ~**1–2 days**.

2. **We maintain templates for him** under the support plan he has already signalled he'll take: *"I'd probably be looking at [being] on that for at least six to 12 months."* **Zero build cost**, small ongoing cost, and he gets exactly what he asks for each time.
3. **Full band designer** — weeks. Not a 20-day item under any reading.

A dedicated session on this was already agreed on Call 2: *"we're probably going to have some questions around this specific piece... we'll probably have a session to flesh that out closer to the time."*

Hold it before build starts. This remains the largest scope risk (Risk 5) — just for a different reason than originally written.

Scope Call 2 — Don't literally embed the app in the Simvoly website

What Matt asked for

"They've got an existing Website, can we 'bolt' this PWA onto their existing website? Shared User Credentials. 'All-in-one' experience. They don't want to re-build the website because it ranks quite high in the SEO space so they don't want to lose traffic."

The goals are all correct. The mechanism is not.

Simvoly is a closed drag-and-drop website builder with **no API, no server-side hooks and no session sharing** (02- \$2). Its pages are client-side rendered. Genuinely embedding an authenticated application inside it means iframes and cross-domain session hacks — fragile, slow, and actively hostile to an offline-first driver app.

What we do instead

Goal	How it's met
Don't lose SEO	The marketing site stays exactly where it is . Not one page touched. Zero SEO risk — strictly lower risk than modifying it
All-in-one experience	App at app.plastago.com.au , matching brand, fonts, colours and logo. Header and footer link both ways. To a user it is one product
Shared credentials	Single sign-on across all surfaces — customer portal, admin console, and both driver builds (native + PWA) . One identity, one login, one place to manage users
Keep the booking page	plastago.com.au/book-a-pickup stays live as a public page and gains two CTAs: <i>"I have an account → Book now"</i> and <i>"New to PlastaGo → Get started"</i>

✅ **RESOLVED.** On the earlier discovery call Matt floated replacing the site (*"hopefully we can replace the website"*, [transcript 09:40]) and Decoded replied *"we probably will rebuild the website."* **That position was superseded.** The later email of **14 Aug 2026** is explicit: *"They **don't want to re-build the website** because it ranks quite high in the SEO space so they don't want to lose traffic."*

The website stays. It is not in scope to rebuild it. That is the cheapest, fastest and lowest-risk path, and it spends **zero of the 14 days** on a marketing site.

⚠️ **Revised after review.** An earlier draft of this recommended **301-redirecting** [/book-a-pickup](https://plastago.com.au/book-a-pickup) to the portal. **That was wrong.** A 301 from a ranking content page to a login screen loses the content

signal, mismatches search intent and raises bounce rate — and worse, it **deletes an inbound lead channel that currently works**. Today new prospects find that page and fill in the form; some of them convert. Keeping the page and branching it preserves both the SEO and the lead. See **M5 · A.1**.

Bonus: it also lets us kill the **dead Cognito form still being served on their homepage (02- \$4)** — a live defect nobody has noticed.

Scope Call 3 — Keep TransVirtual read-only for two weeks after go-live

What the brief implies: build it, switch it on, done.

What we should insist on: a **two-week read-only overlap**.

Why this is the right call

- Their pricing has at least one rule we cannot see: **"Extra Load Time" is auto-generated by TransVirtual** with fractional quantities (06- \$5). There may be others.
- Invoicing runs **daily** at ~7 jobs/day on **7-day payment terms**. There is no quiet window in their calendar and no slack in their cash cycle.
- The cost is trivial — a fortnight of subscription — against the cost of being unable to invoice.

What to say to Matt

"We'll keep TransVirtual switched on but read-only for two weeks after we go live. It costs you one more month's subscription and it means that if anything about your pricing behaves differently to what we've reproduced, you have a fallback and a reference. We'd rather spend a few hundred dollars than risk a week of your invoicing."

5. The eight things that will blow the deadline

Named here so they are managed from day 1 rather than discovered on day 11.

Risk 1 — Pricing must match TransVirtual exactly ⚠️ **HIGHEST**

The risk. Every invoice the new system produces must match what TransVirtual would have produced, to the cent. A pricing engine that is 98% right is a business that cannot bill.

Why it's dangerous. Their rate model is a three-dimensional lookup (**Customer** → **Rate Card** × **Freight Item** × **Zone**) with effective dating, tier fallbacks, per-customer component overrides, two billing methods, nine additional service types (fixed *and* percentage), and fractional quantities. Plenty of room for silent divergence.

Mitigation — the single highest-leverage action in the whole project:

Build a pricing regression harness on Days 6–8 — and never later than Day 10. ⚠️ **It moves later than ideal because days 1–3 are frontend-only (\$13.1).** That is an accepted trade, not an oversight — but it is now the tightest constraint in the plan, for two reasons:

- **MongoDB provides no database-level integrity backstop** (§6A.3), so this harness is the only guard on invoice correctness
- It must be green before migration begins on day 16 Export the last **200–300 completed jobs with their real TransVirtual invoices**. Feed each job through the new engine. **Diff line-by-line** against the real invoice. Target: **100% match** on base price, additional services, GST and total. Every mismatch is either a bug in our engine or an **undocumented rule in theirs** — both must be found in week 1. Keep it in CI permanently so pricing can never silently regress.

Cost: roughly half a day. It is the difference between "we replaced their TMS" and "we broke their invoicing."

Risk 2 — Undocumented rules, chiefly auto-generated "Extra Load Time"

The risk. The Additional Service Approvals queue shows a charge **Created By: System | Qty 3.65 | Extra Load Time | Clarendon Homes**. A fractional quantity generated by the system means **time-on-site beyond an allowance is being billed automatically**, computed from driver timestamps — and we do not know the rule.

✅ **Partly answered by the transcript.** We now know where the duration comes from: Matt described the driver flow as "you hit **Arrived** ... then at the end you hit **Complete Job**. So the time between Arrived and Complete Job, **that's the logged time for that job**." [transcript 30:46] ✅ **Mechanism now identified from the tenant (\$12.2 Q4): it is TransVirtual's demurrage (wait time) feature.** Ask Driver to Confirm Onsite is ON, which locks arrival time and starts the demurrage clock; ClientMobileWarnDriverOnsite15min is ON. The charge is flat-rate \$100, approval-required, single-use per consignment, and **not driver-raisable** — hence *Created By: System*.

⚠️ **Still unknown: the free-time threshold and the unit** (per hour? per 15-min block?). The 3.65 quantity suggests hours. Not present in Global Setup — check the rate card's additional- service override screen next, then ask Matt only if that fails.

Consequence if missed: that revenue **silently stops** at cutover. Nobody notices for weeks because nothing errors — invoices just come out smaller.

Mitigation. Blocking question to Matt today: what triggers it, from which timestamps, what's the free allowance, what's the unit. If unanswered by **Day 3**, ship it as **manual entry** and flag it explicitly in the handover as a known deferral.

Risk 3 — The m²/kg field convention has drifted over three years

The risk. TransVirtual is a freight system with no "area" field, so PlastaGo encode m² into freight dimensions. The invoice screenshot shows **1,800 m²** stored as **Length 1800 × 1 × 1** so that **Cubic = 1800**. But:

- **2026 invoices:** **Weight** = real kg, **Cubic** = m² ✅
- **Feb 2025 jobs:** **Weight** = 699/538/1485 (looks like m²), **Cubic** = 1 ⚠️
- **Mid-2023 jobs:** **Weight** = 0 or 636/800, **Cubic** = 0/1/2/3 ⚠️

The meaning of the same two columns has changed at least twice. A naive migration produces garbage historical reporting and could mis-price.

Mitigation. Resolve the convention **per era and per customer** with Matt before migration. This is also the strongest argument for the new schema having **first-class** **area_m2** and **weight_kg** — see M1.

Risk 4 — Photo migration volume is unknown

The risk. Three years of job photos exist. We do not know how many, how large, or whether TransVirtual can bulk-export them. If it can only be done one job at a time through the UI, 5,079 jobs is not a migration — it's a wall.

Mitigation. **Measure the photo export on Day 1**, before anything else.  **The mechanism is now proven, not just promised:** four POD export rules are **live in production** today (they are what email completion photos to customers), so the export pipeline demonstrably works. What is *not* yet known is **volume and throughput** for a bulk historical pull. Create an image-export rule against a date range and measure it. If a bulk pull is impractical, history becomes **archive-only** — and we learn that on Day 1 rather than in week two. See §12.2 Q2.

Risk 5 — Scope creep through the template designer

The risk. "Custom invoice templates" is the kind of requirement that expands to fill whatever time exists. It is unbounded by nature.

Mitigation. Get **Scope Call 1** agreed in writing before build starts: *resolution and branding now, WYSIWYG authoring in v1.1*. If it isn't written down and agreed, it will consume the sprint.

Risk 6 — Day-21 intake collapse (customer enrolment, M5 Part 2)

The risk. Today the booking form is **public** and the account field is **free text** — anyone can book. The new portal is **authenticated**. If existing site supervisors cannot get in on the first morning, they phone the office. Multiply by every active site and **intake collapses on day one**, regardless of how good the software is.

Why it's dangerous. This failure is **silent**. A supervisor who can't log in doesn't file a bug — they ring the office, or they give up and put the board in a skip. The revenue just doesn't arrive, and the project looks like a failure in week one.

Scale of the problem: 31 accounts active in the last 12 months, with a site-contact population likely in the low hundreds. Large enough to matter, small enough to solve.

Mitigation — all in M5 Part 2:

- **Pre-seed every known user** from TransVirtual contacts + 12 months of Cognito site-contact emails (M5 · C.1)
 - **Invitation campaign at T-5, T-1 and T+1 days**, with activation rate tracked as a **go-live gate** (M5 · C.2)
 - **Personally onboard the top five accounts** — they are 87% of revenue (M5 · C.3)
 - **Unrecognised-booker verification queue for 30 days** — never lose a job because someone couldn't log in (M5 · C.4)
 - **Keep /book-a-pickup public** as a branching page, not a redirect to a login wall (Scope Call 2, revised)
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Risk 7 — RESOLVED BY DECISION: native + PWA driver app

The risk. Matt asked directly [transcript 42:05]: "Will it allow the GPS and stuff — even when the app is not open, if they go back to their home screen, does it still allow the GPS tracking?"

Background geolocation is not reliably available to a Progressive Web App, and on iOS it is effectively unavailable — and Troy is on an iPhone. Continuous location with the app closed requires a **native app**, which means an app-store build and review cycle that does not fit inside 14 days. (It also conflicts with Matt's other requirement: "I don't really want other people to be able to download it" — a PWA distributed by invite link actually satisfies that better than a public store listing.)

Mitigation — this is exactly why M4.2 captures GPS on status events:

	PWA (v1)	Native (v1.1 if needed)
GPS at Acknowledged / En route / Arrived / Complete		
Verifiable arrival time + location per job		
ETA against run sequence		
Live moving dot with the phone in a pocket		

 **DECIDED:** build both, in full.

Flutter for native (proper background location, offline, camera) **and a complete React PWA** alongside. A dedicated mobile developer covers Flutter for the full 20 days; the PWA is built by the web devs off the same Day-3 confirmed design. **Cost and mitigations in \$6A.4** — chiefly: the **offline sync protocol is specified once, in writing**, before either implementation starts.

This closes Q6 and removes the risk — but it creates two new ones, below: **Apple Developer enrolment** and **dual-surface parity**.

Risk 8 — Apple Developer enrolment and iOS distribution

The risk. A native iOS app cannot be sideloaded from a website. Matt does **not** want it publicly listed ("I don't really want other people to be able to download it"). So distribution is TestFlight-internal or Ad Hoc — both of which require an **Apple Developer Program membership (\$99/yr)**, and enrolment involves identity verification that **can take days and is entirely outside our control**.

Route	Review	Public	Notes
TestFlight internal (≤100 testers)	None	No	 Best fit.  Builds expire after 90 days — someone must re-upload quarterly
Ad Hoc (≤100 devices/yr)	None	No	 Works. Each device registered by UDID; new phone = rebuild
App Store	Full	Yes	 Matt doesn't want it public

Android is simpler: ship the APK from their own site. Play Protect warns on install, and there is **no auto-update** unless we build an update check.

Mitigation:

- **Start the Apple enrolment on Day 1**, before any code. It is the longest external lead time in the project and it costs nothing to have it ready early.
 - **Ship the PWA first** so the drivers are never blocked waiting on Apple.
 - Decide TestFlight vs Ad Hoc early — TestFlight is easier to distribute, Ad Hoc avoids the 90-day expiry treadmill.
 - **Tell Matt about the 90-day TestFlight expiry now**, not in month four.
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Risk 9 — AI PO extraction accuracy (I8 + I6)

The risk. The automated PO-from-email pipeline is the client's *stated core pain* — the project overview opens with "automating the receipt of purchase orders." It is also the only genuinely probabilistic component in the build. Builders' POs arrive as PDFs, scans, photos of paper and email bodies, in dozens of layouts.

Why it's dangerous. An extraction pipeline that is 85% accurate and **silently wrong** on the rest is worse than manual entry, because nobody checks it. A wrong PO number means an invoice that bounces in the builder's AP system — the exact problem we were hired to fix.

Mitigation — design for the failure case first:

1. **Confidence thresholds with a human queue.** Above threshold, auto-create; below, route to an office review queue with the source document side-by-side. **Never silently guess.**
 2. **Extract, then match.** Match the extracted account/site/reference against existing records rather than trusting free text. A PO that can't be matched goes to the queue.
 3. **Keep manual PO entry (M2.10) as a first-class path**, not a fallback. It must work on day one regardless of the AI.
 4. **Measure it.** Log extraction confidence and correction rate from the start so accuracy is a number, not an opinion.
 5. **Test against real POs early** — ask Matt for 20–30 recent PO emails in week 1.
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6. The modules — what we are actually building

Ten modules. Each section states **what it is**, **who uses it**, **why it matters**, then lists its feature items with a description and a concrete example.

M1 · Platform Foundation

What it is. The data model, identity system, permissions and audit trail that everything else sits on. Not a screen — the thing that makes the other nine modules possible.

Who uses it. Nobody directly. Everybody indirectly.

Why it matters. Four decisions here are effectively irreversible after day 2. Getting any of them wrong means a rewrite in month three, not a patch.

Feature items

1.1 — Multi-brand model (PlastaGo / EasyLift / BrickGo) Brand is a first-class dimension on accounts, jobs, invoices, templates, email senders and logos — not a setting.

Why now, not later: this is **already real in production**. TransVirtual holds **EasyLift Recycling Invoice (KG)** and **(m2)** templates, and driver Troy Holm logs in as **troy@easylift**. BrickGo was registered under the same ABN in April 2026. Building single-brand and retrofitting is a rewrite of the invoicing layer. **Example:** a job for an EasyLift customer produces an invoice with the EasyLift logo, EasyLift ABN line and EasyLift email sender, while a PlastaGo job on the same day produces the PlastaGo version — automatically, with no user choice involved.

1.2 — The party model: Account ≠ Builder ≠ Site ≠ Contact Four distinct entities, not one "customer".

Proven by production data: *iPlasta Pty Ltd* is the **account** that gets invoiced; *Fowler Homes, GJ Gardner* and *King Homes* are the **builders** whose sites are being serviced. *Durnco Group Pty Ltd* is the account; *Mirvac* and *Sharwood* are the builders. **Example:** iPlasta books a pickup at a GJ Gardner site in Oran Park. The invoice goes to iPlasta on iPlasta's rate card; the job record shows GJ Gardner as the on-site builder; the site supervisor who gets the SMS is GJ Gardner's; and the Oran Park site accumulates its own history across 14 pickups over the life of the build.

1.3 — area_m2 and weight_kg as first-class fields Dedicated, correctly-typed, correctly-named columns.

Why: TransVirtual has neither, so PlastaGo overload freight dimensions and the meaning has drifted twice (Risk 3). **Example:** a 1,800 m² job is stored as **area_m2 = 1800**, not as **length = 1800, width = 1, height = 1**.

1.4 — Job numbering continues the existing sequence New jobs start where TransVirtual left off (currently ~61,300).

Why: three years of consignment numbers are referenced in builders' AP systems, on paid invoices, and in every conversation with Clarendon and Domaine. Restarting at 1 breaks continuity with the people who owe them money. Invoice numbers likewise continue from ~104,100.

1.5 — Role model and permissions Seven roles, mirroring their existing TransVirtual security groups so nothing has to be re-learned: **SuperAdmin · Operations · OfficeStaff · Allocator · Driver · Customer-Administrator · Customer-Site Supervisor**.

Example: a Customer-Site Supervisor at GJ Gardner can book and view their own site's jobs but cannot see pricing, other sites, or any other builder's work. Their Customer-Administrator can see all of their company's sites, jobs and invoices.

1.6 — Audit log on every state change Who changed what, when, from what value to what value — immutable.

Why: today a shared mailbox and shared page password mean **zero accountability**. **Example:** "Who changed job 61402's ready date from 12 Aug to 19 Aug?" → answerable in one click, with a name and a timestamp.

1.7 — Retention & data residency 7-year retention floor (their Privacy Policy commits to 7; RRO14 requires 6), Australian data residency, Australia/Sydney timezone normalisation.

Note: TransVirtual currently renders their timestamps in **NZST/NZDT**. All migrated timestamps must be converted, with daylight saving handled correctly.

Workflows covered: W1, W2, W3, W16, W17, W71 (partial)

M2 · Admin & Office Console

What it is. The desktop web application the PlastaGo office team lives in. It replaces TransVirtual's day-to-day screens — the ones they pinned as favourites.

Who uses it. Matthew Browne (Operations), office staff, admin.

Why it matters. This is the **parity module**. If this isn't at least as good as what they have, nothing else matters — they won't switch. Everything else is upside; this is the price of entry.

Feature items

2.1 — Create a Job (*their #1 daily screen*) A single form to raise a pickup: account, builder, site, customer reference/PO, ready date, service level, freight item, expected m², bag count, notes, photos.

Improvement over today: account and site are **pickers, not free text**; the address is **geocoded with a map pin**; and the rate card resolves and shows the **expected price before saving**. **Example:** office receives a phone booking from iPlasta. They type "iPI" → select the account → pick an existing site or drop a new pin → enter the customer reference → set ready date → enter 855 m² → save. The screen shows "**Estimated \$356.80 ex GST — Sydney zone, \$220 service + 855 m² × \$0.16**" before they commit. (*That figure is real: it is invoice 104071, reproduced to the cent by the published rate — see 07-\$5A.4.*)

2.2 — Job List (*their #2 daily screen*) — F6 Searchable, filterable list of past, present and future jobs: by status, account, builder, driver, date range, suburb, zone, invoice state. Every job carries a **unique job number** continuing the existing sequence.

Example: "Show me every unallocated job with a ready date in the next 3 days in the Newcastle zone" → one filter set, immediately actionable.

2.3 — Job detail record — F4, F5, F7 The full record: all parties, the site, the timeline of status changes with timestamps and actors, **m² and weight**, itemised charges, photos, documents, comments, exceptions and linked invoices.

Per-customer capture config: some accounts are **m²-only** (iPlasta, Fornari, Wisdom all record weight = 0); others capture **both** (Clarendon, Domaine). The form adapts. **Example:** open job 61328 and see: booked 3 Jul by *Domaine*, allocated to *Troy Holm*, arrived 09:14, marked **futile** at 09:21 with reason and two photos, \$120 futile charge raised, awaiting office review.

2.4 — Cancel & reschedule — F38 Cancel with a reason code, or move a job to a new date, with the customer notified automatically and the whole change audited.

Business rule from their own spec: a customer may self-serve a date change **only if the job is not already on a run sheet**. After that it becomes an office action.

2.4a — The SLA clock NEW `target_date = customer's ready_date + 5 business days.`

✓ **Corrected by Call 2.** The website says "3 to 5 working days" *from the request*. The operative rule is different: "*they tell us they're ready from [a date] and **we need to pick them up within five business days of that time***" [C2 07:17]. **Most jobs have no time**, only a date window. A **minority have a hard time window** — e.g. a traffic-control permit valid "*up until 12 o'clock*". Model both.

2.5 — Delay and exception reason tracking — F30, F54 Structured reason codes (not free text) on every delay, futile, cancellation and contamination, so they become reportable.

Example: end of month shows "*14 delays: 6 × site not ready, 4 × access blocked, 2 × crane unavailable, 2 × weather*" — actionable, and chargeable where the customer certified otherwise.

2.6 — Futile Pickup Review queue (*their #4 daily screen*) Every driver-marked futile pickup lands here with its reason, photos, GPS and timestamp. Office reschedules or cancels — **either way the \$120 futile fee applies**, exactly as Matt described.

The critical change: this queue **chases**. Ageing badges, a dashboard counter, and a daily digest email. Today TransVirtual has this exact queue and **one entry has sat unactioned since 28 August 2025** — a year, at \$120.

2.7 — Additional Service Approvals queue (*their #3 daily screen*) Charges raised by drivers (contamination, extra load time) queue for office approval before they can be invoiced, with the driver's photo evidence attached.

Why it matters commercially: ~2 contamination charges per week at \$90 ≈ **\$9k/year** flows through this queue. Several are currently weeks old and unapproved. **Example:** Troy marks the Domaine job contaminated and photographs the timber offcuts in the bag. Office opens the queue, sees the photo, approves. A \$90 line is added and — if Domaine requires a PO — the charge moves to the *Awaiting PO* queue rather than blocking the main invoice.

2.8 — Customer account management — F29 Create and maintain accounts: customer code, ABN, brand, rate card assignment, PO policy, capture configuration (m²-only vs m²+kg), payment terms, contacts and their notification preferences, sites.

Example: setting Wisdom Properties' account applies the *Wisdom* rate card, attaches their bespoke "*Fuel Levy – Wisdom*" (\$20) additional service, and marks them m²-only.

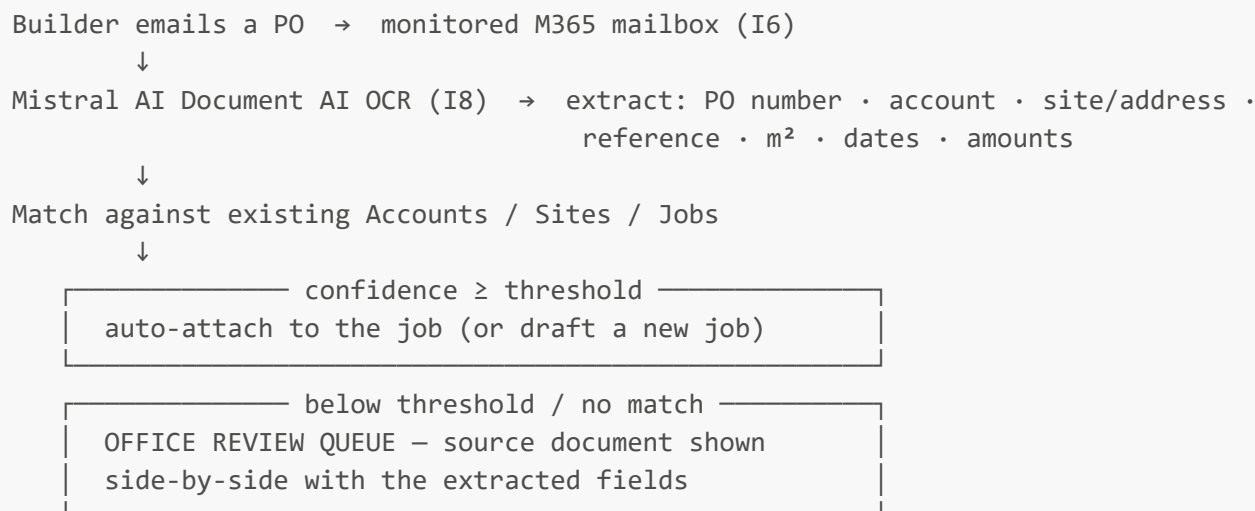
2.9 — Site / project register Reusable geocoded sites with lot number, access notes, gate hours, induction requirements and crane availability, so they're typed once, not once per booking.

Why: their booking form's placeholder literally pleads "*Please use both Lot and Street Number where possible*" — because drivers get lost in greenfield estates where street numbers don't exist yet. Real examples from their data: **Lot 1097 (#46) Allambie Circuit, Lot 3141 Pilaster Street.**

2.10 — PO capture, attachment and enforcement — W46, W48 Record a PO number against a job, attach the PO document, and enforce PO presence for accounts configured to require one. **Manual entry remains a first-class path** — it must work regardless of the AI pipeline.

2.12 — ★ Automated PO ingestion from email NEW — I6 + I8, W46

📌 **Moved into scope by the 20-day plan.** This is the client's **stated core pain** — the project overview opens with *"automating the receipt of purchase orders, job allocations and driver notifications."*



Design rules, driven by Risk 9:

- **Never silently guess.** Below-threshold extractions go to a human queue, never straight through.
- **Extract, then *match*.** Resolve to real Account/Site records rather than trusting free text — an unmatchable PO is a queue item, not a new record.
- **Log confidence and correction rate** from day one, so accuracy is a measured number.
- **Manual entry (2.10) always available**, and the system must be fully usable with the pipeline switched off.
-  **Ask Matt for 20–30 real PO emails in week 1** to tune against actual layouts.

Example: Clarendon emails a PO PDF at 4:47pm referencing *Lot 77 Britannia Road*. The pipeline extracts PO **29916613/096**, matches the site to an existing Clarendon job awaiting a PO, attaches it, and releases the invoice — with no human touch. A scanned photo of a handwritten PO from a small builder scores low confidence and lands in the queue for a 10-second confirmation instead.

2.11 — Job documentation & comments — W49, W56, W68 Attach documents to a job; internal and customer-visible comment threads.

Workflows covered: W8, W45, W46*, W47, W48, W49, W51, W52, W55, W56, W61, W62, W65, W67, W68, W120, W122 *(*partial*)

M3 · Allocation & Dispatch

What it is. The board where jobs get assigned to drivers and days, and run sheets get produced.

Who uses it. The Allocator / Driver Manager (in practice, Matt or office staff).

Why it matters. With transport bundled into the price, every wasted movement is straight margin loss. But with only **2 active drivers and ~7 jobs/day**, this needs to be *fast and clear*, not clever.

Feature items

3.1 — Allocation board — W100, W101, W104 Drag-and-drop or click-to-assign view: unallocated jobs on one side, drivers × days on the other. Reassign freely; the driver is notified instantly.

Deliberately manual. Automatic assignment (F58/W109) is **out** — with two drivers, manual allocation is faster, safer and more transparent than a rules engine. Revisit when the fleet grows.

3.2 — Run sheets — F33, W100 The day's ordered job list per driver, viewable in the office and on the driver's phone, printable, with addresses, references, contacts and notes.

Example: Troy's Tuesday run sheet: 7 jobs, Oran Park → Catherine Field → Gledswood Hills, with lot numbers, site contacts and expected m² per stop.

3.3 — Geocoding and map view — I3 (Google Maps) Every site geocoded with a confirmed pin; jobs plotted on a map so the allocator can cluster by geography visually.

Scoped honestly: this is **visual clustering to support human decisions**, *not* automated route optimisation (F19), which is v1.1. A map that shows six jobs in Oran Park and one in Newcastle is 80% of the value at 5% of the cost.

3.4 — Driver availability view — W111 (simplified) A simple per-driver capacity/availability column on the board — who's working, who's off, how many jobs they already have.

Not the full driver-availability dashboard (F40), which is v1.1. With two drivers a column is sufficient.

3.5 — Delay tracking — W108 Jobs approaching or breaching their target date are highlighted on the board.

Workflows covered: W100, W101, W102, W104, W108, W111*, W120, W122

M4 · Driver App

What it is. An **offline-first driver app shipped as BOTH a native build and a PWA**. Replaces the TransVirtual mobile app (currently v1.6.4 on Troy's iPhone and James's Samsung S23 Ultra). **One dedicated mobile developer for the full 20 days.**

	Flutter (native)	React PWA
Background GPS (phone in a pocket)	✅ The reason it exists	❌
Foreground GPS on status events	✅	✅
Offline queue, camera, photos	✅	✅
Install friction	TestFlight / APK	Just a link
Iteration speed	Build + distribute	Instant
Distribution	iOS: TestFlight or Ad Hoc · Android: APK from their site	Any browser, no install

Approved build approach: two implementations, one contract. **Flutter** for the native app (Drift local DB, `flutter_background_geolocation`) and **React + Vite + vite-plugin-pwa** for the PWA. They share **no code** — they share the **OpenAPI contract** and a **written offline sync protocol** (§6A.4, §6A.9).

Ship the PWA first. Drivers are then never blocked waiting on Apple enrolment (Risk 8), and Flutter lands as the upgrade rather than a dependency. ⚠️ **Every driver feature is now a two-place change.** Keep a parity checklist (§6A.4).

Who uses it. Two drivers today. It must be excellent for them specifically.

Why it matters. This is where **evidence is captured** — and evidence is what turns disputed charges into paid charges. It is also where every data-entry error becomes a billing error.

Design constraints (from §5- §5): drivers work outdoors, one-handed, in gloves, in sun glare, in areas with poor mobile coverage. Big targets, few taps, **must work with no signal** and sync when it returns. A driver app that fails offline is worse than paper.

Feature items

4.1 — My run sheet & job details — W35, W31, W38 Today's jobs in order, with everything needed on site: address with lot number, map/navigate button, site contact with tap-to-call, customer reference, notes, expected m², access notes.

Example: Troy taps job 3 of 7, sees "Lot 328 / 69 Horologium Road, Austral — gate code 4417, crane available 7–11am, contact Dave 04xx" and taps Navigate.

4.2 — Status updates & GPS — F45, F11, W26, W29, W36 Each status change captures a timestamp and GPS position automatically.

↑ **F11 is now FULL, not lite.** With the native shell, **continuous background location** is available — so the office gets a live driver map, and customers can get a real ETA (F26/M5.7). The PWA still captures location at each status event as the fallback.

The six statuses actually used in production, across all 5,079 jobs (§6- §2): **Assigned** → **DriverPickupAcknowledge** → **InTransit** → **Completed** → **AdministrationComplete**, plus **PickupFutileCompleted** as the exception branch. **We are not building a 20-state machine nobody uses.**

+ **We add one: Arrived / On Site.** TransVirtual has no such status, but Matt's workflow depends on it — it is what triggers the Site Risk Assessment (M4.8) and it starts the job clock: "*the time between Arrived and Complete Job, that's the logged time for that job*" [transcript 30:46]. That duration is almost certainly the basis of the **Extra Load Time** charge (Risk 2).

4.3 — Capture m² and weight — F34, W30, W44 Simple numeric entry, large keypad, per-customer configuration so drivers only see the fields that account requires.

Confirmed by the client's own spec: "*real-time weight measurement during pickups (simple data entry of weights)*" — no scale integration, no hardware.

These are two different quantities, not two units (§7- §5A):

- **area_m2 = the size of the job** — board *installed* in the house, usually known before the pickup from the builder's order. **This is what gets priced.**
- **weight_kg = the waste actually recovered.** Measured on the day. Averages ~6.8% of the installed board weight in their real data.

Per-customer capture config: iPlasta, Fornari and Wisdom record **m² only**; Clarendon and Domaine record **both**.

4.4 — Tip-off weight & end-of-run reconciliation  — *the Allocator's "Review tip-off weights" workflow* At the end of a run the driver tips the whole load off at the facility and records the **tip-off weight** — the weighbridge or docket figure — with a photo of the docket.

✓ **Algorithm confirmed by Matt** [[transcript 14:20–17:45](#)]. **Not every job can be weighed** — that is the whole reason this exists:

- **~60–70% of jobs are bagged** → weighed on a **crane scale**, driver types it in
- **~30%+ are hand-loaded** → **cannot be weighed at all**; there is no bag to lift

tip_off_total_kg	weighbridge, per run
- Σ crane_scale_weight (bagged jobs)	actually measured
= remainder_kg	
÷ count(hand-load jobs on that run)	
= imputed_weight_kg	per hand-load job

Matt's own example: 5 jobs, 3 bagged @ 200 kg, tip-off 846 kg → $846 - 600 = 246 \div 2$ hand-load jobs = 123 kg each.

Purpose — explicitly NOT billing: *"This isn't really for invoicing. This is more for us to understand on the back end what the average weight is for our **costing**."* Plus audit: *"when I get the **bill at the end of the month for all the tip-offs**, I can audit that against what we recorded at the time."*

What it feeds — note that billing is *not* the main driver:

1. **Diversion certificates (F52 / M9.5)** — the tonnes-diverted figure on the certificate **is** this number. It goes into builders' Green Star submissions and must be defensible
2. **Mass balance** — evidences the "100% recycling success rate" claim, currently unevidenced
3. **EPA RRO14** — quantity records, 6-year retention ([04-](#) \$2.7)
4. **Driver accountability** — persistent variance is an operational signal
5. **Billing** — ✓ **explicitly not billing**. Matt confirmed weight is estimated *from* m² for pricing; the measured kg exists to make that estimate better. This usefully de-risks the whole feature

Deliberately simple in v1: driver enters the tip-off figure + docket photo; the system applies the deduct-and-average rule; the Allocator gets a review queue with the variance. **No weighbridge hardware integration.**

 **New question this raised:** they receive a **monthly bill for tipping**, which means tipping is a *purchased service* — sitting awkwardly against the website's "*our own recycling facility*". Whose site is it? See [09-](#) \$1.

4.5 — Photos — F7, F14, W28, W34 Multi-photo capture, automatically timestamped, geotagged and attached to the job. Queued locally and uploaded when signal returns. **No limit on count** — Matt was explicit.

Their actual photo protocol [transcript 20:15–21:38]:

1. Front of the site · 2. The pile **before** · 3. The pile **after** ·
2. The site **closed** · 5. *If it can't be closed* — **the cars still on site**

That last one is pure commercial defence: "*a lot of the time we get **blamed for leaving everything open**, so we take evidential proof that this person was still here when we left.*"

Typical 10 photos per job, sometimes 18–20.

NEW **Configurable required-photo prompts** — Matt asked for "*a prompt of [what's] in the photos that are required, and a space to put in any others.*" So: a per-site or per-job checklist of required shots, plus free-form extras. Cheap to build, and it standardises the evidence that defends their charges.

4.6 — Mark Futile + reason — W27 One prominent button. Choose a structured reason (job not ready / no truck access / site closed / crane unavailable / nobody on site), add photos, submit. Job routes to the office review queue and a **\$120 futile fee** is raised.

This is the money loop. The customer certified at booking that the job was ready and accessible (03-§3). Photo + GPS + timestamp at the point of failure turns a disputed phone call into an invoice line that survives challenge.

4.7 — Mark Contaminated — W27, W32 Flag contamination with type, estimated extent and mandatory photo evidence. Raises a **\$90 contamination charge** into the approval queue.

Improvement over today: currently it's a binary flag. We capture *what* the contamination was and *how much*, so it becomes reportable and defensible — and so repeat-offender sites can be identified.

4.8 — Site Risk Assessment, SWMS & pre-start — F56, F14, W37, W41 ⚠️ *bigger than it looks* Three related things:

(a) Driver Pre-Start Checklist — port their existing TransVirtual mobile form.

Chain of Responsibility under the Heavy Vehicle National Law makes pre-start checks an operator obligation, not just a driver one (04- §2.10).

(b) Site-Specific Risk Assessment + SWMS → builder's portal — a **5-step workflow**, not a checklist. Matt described it precisely [transcript 28:16–30:14]:

```
Driver taps ARRIVED
↓ Site Risk Assessment form POPS UP automatically
↓ Driver completes it on the spot
↓ Generate PDF:  page 1 = risk assessment
                  page 2 = standard SWMS (versioned, "updated yearly")
↓ Driver SCANS THE QR CODE ON THE SITE FENCE
↓ Uploads the PDF to the BUILDER'S OWN PORTAL
↓ Job proceeds
```

Required by some clients for every site, before the driver may start. Scope reality: form engine + PDF generation + PDF merge + camera QR scanning + handoff to an external URL — and it must work

offline, because the driver is standing at a fence. This is several features, and it is the largest single item in M4.

(c) **Ad-hoc site hazard logging** with photos.

4.9 — Vehicle defect reporting — F43, W33 Driver logs a vehicle problem with a photo; it notifies the office and lands against the vehicle record in M9.7.

↑ **Now part of the full F43 feature**, not a stub — see M9.7.

4.10 — Job notes on completion — F41 now OUT of scope

F41 (structured driver feedback flagged for office review) has been dropped. A plain free-text note field on the job remains — drivers can still record *"bag was half buried in mud, needed 20 extra minutes"* — but there is no separate review-and-triage flow. The **Extra Load Time** charge is system-generated from on-site duration anyway (M6.7), so the commercial outcome is unaffected.

4.11 — Push notifications — F45, W36 New job assigned, job cancelled, job details changed, run sheet updated.

4.12 — Offline-first sync Everything above works with no signal. Actions queue locally, sync automatically, and the driver can see clearly what has and hasn't synced.

Workflows covered: W25, W26, W27, W28, W29, W30, W31, W32, W33*, W34, W35, W36, W37, W38, W41, W44 — plus *Allocator · Review tip-off weights* (unnumbered in the brief), via M4.4

M5 · Customer Portal & Enrolment

What it is. A **mobile-responsive Progressive Web App** for PlastaGo's customers. Two levels of access: **Customer Administrator** (head office — sees everything for their company) and **Site Supervisor** (sees only their own sites).

This is what F21 means. Not a native app — a portal that works beautifully on a phone. See §2.1.

Who uses it. Site supervisors at builders and plastering contractors; head-office admins at iPlasta, Domaine, Clarendon, Fornari, Wisdom.

Why it matters — the commercial argument. Five accounts are 87% of revenue; one is 34%. Losing iPlasta removes a third of the business. TransVirtual gives those customers **nothing** — today their entire digital experience is a public tracking URL printed on a PDF invoice. REGYP, the 19-year-old market leader, offers nothing either (04- §3).

This module is not an efficiency feature. It is a retention weapon aimed at five named companies, and it is the single biggest competitive differentiator in the build.

How this module is organised. Two parts:

- **Part 1 — Using the portal** (items 5.1–5.15): what a logged-in customer can do.
- **Part 2 — Getting in** (Journeys **A / B / C**): how a human ends up able to log in at all, plus the safety net for when none of the routes work. (*Previously module M11.*)

Part 1 — Using the portal

5.1 — Book a Pickup — F9, F21, W88 Authenticated booking with saved sites. Roughly **four fields** instead of today's seventeen, because the system already knows who they are, what their sites are and what their rate is.

Before: 17 fields, 13 mandatory, account name typed by hand on a building site in the rain, no validation (03-). **After:** pick site → ready date → expected m² → bags → confirm. Account, builder, address, contact and certifications all pre-filled and verifiable.

5.2 — Job readiness certification — W86 The three certifications (job ready / truck accessible / free of contaminants) captured against a **named, authenticated user** with a timestamp — instead of a typed name in a text box.

Why this matters: when the driver arrives to a job that isn't ready, the futile charge is backed by *"David Chen, GJ Gardner, certified on 12 Aug at 14:32 that this job would be ready"* — not by an unverified string.

5.3 — Site access information — W87, W98 Supervisors maintain their own site access notes, gate hours, induction requirements, crane windows and contacts.

Directly prevents futile pickups — which are a cost to PlastaGo and an annoyance to the customer, so both sides are motivated.

5.4 — Edit / reschedule a pickup — F38, W97 Change the ready date **while the job is not yet on a run sheet**; after that, request a change which routes to the office.

This rule is taken verbatim from the client's own site map.

5.5 — Request urgent pickup / change urgency — F28, W89, W80 Flag a job as urgent; office is alerted and the job is highlighted on the allocation board.

5.6 — Preferred pickup times — F46, W92 Specify preferred windows or blackout times per site or per job.

5.7 — Live job status & dashboard — F26, W95, W38 See exactly where every job is: booked, allocated, driver on the way, on site, completed.

Replaces: phoning the office to ask. Every one of those calls is a self-inflicted cost.

5.8 — Job history — F18, W77, W83, W99 Every past job with full detail, searchable and filterable by site, date and reference.

5.9 — Completed jobs with photos — W99 The full completion record: m², weight, driver, times, and **all site photos**.

Beats today's experience, which is a mydat.info link printed on a PDF.

5.10 — Invoices — F8, W72, W73 View and download outstanding and paid invoices as PDF, with PO number and job reference.

5.11 — Monthly reports — F1, W74, W76, W78, W82 Self-serve monthly pickup volume reports per company and per site — the report PlastaGo currently produces and sends manually.

5.12 — Recycling certificates — F52, W84 Download a Certificate of Recycling for a project, a period or a job. See M9.5.

5.13 — Upload job documents — F32 now OUT of scope

Customers email documents to the office as they do today; the office attaches them via M2.11. Revisit in v1.1 — the upload pipeline exists, so it is a small add later.

5.14 — Site Supervisor management — W70 Customer Administrators add, remove and manage their own site supervisors, and control which sites each can see.

Removes PlastaGo from the loop entirely on a task that currently means a phone call.

5.15 — Account preferences — F29, W71, W81 Update company details, contacts and notification preferences.

Workflows covered by Part 1: W70, W71, W72, W73, W74, W75, W76, W77, W78, W80, W81, W82, W83, W84*, W85, W86, W87, W88, W89, W91, W92, W95, W97, W98, W99

Part 2 — Getting in: enrolment & onboarding

Merged from what was previously M11. Enrolment is the front door to this portal, so it belongs here rather than in a module of its own.

Why it exists: the original scope defined an authenticated portal without defining **how anyone gets into it**. Today PlastaGo's front door is wide open — the Cognito form is public and "Your Account" is a free-text box. Replacing that with a login wall, without designing the way in, would **silently strangle intake on day 21**.

This part is **not optional**, and §7 explains what it displaces to pay for it. Journey items keep their **A.x / B.x / C.x** labels (referenced elsewhere as **M5 · A.1** etc.).

⚠️ **Two of these three journeys are not literally "portal" screens** — say so plainly when planning the build:

- **Journey A** (lead capture → onboarding queue → convert to account) is built in the **admin console (M2)**; only the public enquiry form is customer-facing.
- **Journey C** (pre-seeding users, invitation campaign) is executed as part of **migration & cutover (M10)**.
- **Journey B** is the only one that is purely portal.

Added after review. The original scope defined an authenticated portal without defining **how anyone gets into it**. Today PlastaGo's front door is wide open — the Cognito form is public and "Your Account" is a free-text box. Replacing that with a login wall, without designing the way in, would **silently strangle intake on day 15**.

This module is **not optional**, and §7 explains what it displaces to pay for it.

What Part 2 is. The three distinct paths by which a human ends up able to book a pickup — plus the safety net for when none of them work.

Who uses Part 2. Prospects, existing customers' site supervisors, PlastaGo office staff.

Why Part 2 matters. Intake is the top of the revenue funnel. Every point of friction here is either a lost job or a phone call to the office — and reducing phone calls is the outcome Matt will actually judge the system by.

The three journeys — they are genuinely different problems

	Journey	Volume	Right answer
A	New company → new account	~ 1.6/month (53 accounts in 38 months)	Human-mediated. It's a sales decision — rate negotiation, credit, terms. Do <i>not</i> automate
B	New site supervisor at an existing account	High and continuous — construction site turnover	Frictionless self-enrolment. This is where a login wall does the damage
C	Existing users at cutover	31 active accounts , one-time, Day 20	Pre-seeded and invited before go-live — a migration task, not a feature

Journey A — New company becomes a customer

Today: they either phone 1300 395 438, or they fill in the public form with a made-up account name and the office works out who they are. That second path means **leads arrive disguised as jobs** — and some of them convert.

The trap to avoid: if we simply lock the form behind a login, we don't just block bad bookings — **we delete an inbound lead channel that currently works**, on the page that ranks in Google. That would be a real commercial regression.

A.1 — Keep `/book-a-pickup` public, and make it a branch The page stays exactly where it is on Simvolvy, keeps its content and its SEO, and the dead Cognito iframe is replaced with two clear paths:

- **"I have an account"** → **Book now** (deep link into the portal)
- **"New to PlastaGo"** → **Get started** (public enquiry form)

See Scope Call 2 — an earlier draft proposed a 301 redirect to the portal; keeping the page and branching it is strictly better for SEO *and* conversion, and preserves the lead.

A.2 — Public enquiry form → creates a Lead, not a Job Company, contact, phone, email, typical volume, suburbs/region, expected frequency, how they heard about PlastaGo. It creates a **Lead** record — deliberately *not* a job, because nothing can be priced or scheduled until an account exists.

Example: a new plastering contractor in Newcastle submits an enquiry at 9pm. It lands as a Lead, auto-acknowledged by email, and appears in the office's onboarding queue next morning with the Newcastle zone already inferred from the postcode.

A.3 — Onboarding queue in the admin console Leads with status (new / contacted / quoted / won / lost), owner, and notes.

This is the first CRM capability PlastaGo has ever had — and it's nearly free, because it's one more queue in a console that already has three. It also finally makes *"how many enquiries did we get and how*

many converted?" answerable.

A.4 — Convert Lead → Account wizard One flow: create the account, assign a rate card (named / Tier 1–4 / default), set PO policy, capture configuration (m²-only vs m²+kg), payment terms, brand, contacts, first site — then **send the invitation**.

Example: Matt qualifies the Newcastle lead on the phone, agrees Tier 2 rates, clicks Convert, and the customer receives a welcome email with a link and their customer code before he's hung up.

A.5 — One-off / prepaid jobs ⚠️ *needs a decision from Matt* Their data contains a **"PrePaid Customer"** account (5 jobs) and Recycling Near You lists them as accepting **residential quantities**. There is a genuine casual segment — the long tail shows ~26 accounts with fewer than 10 jobs ever.

v1 approach: the public enquiry form handles these; the office books them against the existing "PrePaid Customer" account, as they do now. **No self-service one-off booking in v1** — it's 0.1% of volume and it needs payment handling. **This is the one place Stripe (I4) might genuinely earn its place** — see §8, where I've revised the reasoning behind dropping it.

Journey B — New site supervisor at an existing account ★ *the one that matters*

This is the high-volume, high-risk journey. Construction site supervisors change constantly. If getting access is slower than the current "type your company name into a public form", **they will phone the office instead and we will have made the problem worse.**

Three routes in, in order of how we expect them to be used:

B.1 — Site booking link (*expected to be the dominant route*) Every Site carries a shareable link. Head office or PlastaGo sends it; the supervisor taps it, enters their mobile, receives an **SMS one-time code**, and can immediately book **for that site only**.

Why this fits construction: links get shared in WhatsApp groups and site inductions. No app, no password, no invitation admin, no waiting. **Example:** Clarendon's contracts administrator pastes the Castle Hill site link into the site WhatsApp group. The new supervisor taps it, enters 04xx xxx xxx, types the 6-digit code, and books a pickup — total elapsed time under a minute, on a phone, on site.

B.2 — Join by customer code They already have customer codes — **IPL001, DOM001**. A supervisor enters the code plus their mobile, verifies by SMS, and is attached to that account as a Site Supervisor.

Industry-familiar: competitor REGYP already requires a **"Customer Code"** to deposit waste, so builders understand the concept. **Safety valve:** new joins can be set to require approval (by the Customer Administrator or PlastaGo) — configurable per account, because Clarendon may want control while a small contractor won't care.

B.3 — Invitation by the Customer Administrator — already in scope as **M5.14 / W70** Head office adds supervisors directly and controls which sites each can see.

Best for organised accounts (Clarendon, Domaine). Not sufficient on its own, because it depends on head office being responsive — which, at 4pm on a Friday, they aren't.

B.4 — Email-domain auto-association (*optional, per account*) If a signup email matches a domain whitelisted on the account (@clarendonhomes.com.au), auto-attach with Site Supervisor role.

Caveat, worth stating: many site supervisors and subcontractors use personal Gmail addresses or only have a mobile. **This is a convenience, never the primary route** — which is exactly why B.1 and B.2 are mobile-first.

Journey C — Existing users at cutover ⚠️ *this is a go-live risk, not a feature*

The failure mode: Day 15, a Domaine supervisor goes to book a pickup, can't get in, and phones the office. Multiply by every site. **Intake collapses on the first morning and the project looks like a failure regardless of how good the software is.**

C.1 — Pre-seed every known user before go-live Build the user list from two sources:

- **TransVirtual** — account contacts, and every distinct site-contact email on the last 12 months of consignments
- **Cognito Forms** — every "Site Contact Email" submitted in the last 12 months

Scale is manageable: 31 active accounts, and the site-contact population is likely in the low hundreds. This is a query and a mail merge, not a project.

C.2 — Pre-cutover invitation campaign T-5 days: email + SMS to every seeded user — "*PlastaGo is moving to a new booking portal on [date]. Here's your link. Nothing to install.*" T-1 day and T+1 day: reminders to anyone who hasn't activated.

Track activation rate as a **go-live readiness metric**. If the top five accounts aren't activated, don't cut over.

C.3 — Personally onboard the top five accounts iPlasta, Domaine, Clarendon, Fornari and Wisdom are **87% of revenue**. These get a phone call and a walkthrough, not an email.

This is also the best sales moment of the whole project — it's when they see the portal for the first time and realise nobody else in the industry offers it.

C.4 — The safety net: unrecognised-booker verification queue 🚫 For the **first 30 days**, someone who reaches the portal without a recognised account can still submit a booking. It lands in a **verification queue** with a warning banner rather than being rejected.

Design principle: never lose a job because someone couldn't log in. An authenticated portal's characteristic failure is *silent* lost revenue — the customer doesn't complain, they just phone a competitor or throw the board in a skip. The queue makes that failure loud and recoverable. **Example:** a supervisor at a new GJ Gardner site books without a code. The job appears flagged "unverified booker"; the office recognises the site, links it to iPlasta's account, and the job proceeds normally. The supervisor is enrolled automatically as a side effect.

C.5 — Office can always book on behalf Already in scope (**M2.1**). The phone line does not go away, and shouldn't — some supervisors will always ring. The difference is the office now types it into a system that prices, schedules, notifies and invoices automatically.

What we are deliberately NOT doing

Not doing	Why
Self-service account creation with instant credit	Rates are negotiated and credit is extended. That's a business decision, ~1.6 times a month. Automating it adds risk and saves nobody time
Passwords	Site supervisors will not remember them. OTP only — email for office, SMS for site
Forcing head office to provision every supervisor	Depends on their responsiveness; would push people back to the phone
Keeping the Cognito form live "just in case"	Two intake paths = two sources of truth = the exact re-keying problem we're eliminating. The verification queue (C.4) is the safety net instead

Workflows covered by Part 2: W8 (extended), W52, W68, W70, W71, W75, W81, W85 — plus the previously-unmapped enrolment need behind the whole portal

M6 · Pricing & Rate Card Engine

What it is. The engine that turns a completed job into a correctly-priced set of charge lines. Invisible to users; the most consequential code in the system.

Who uses it. Nobody directly. It runs on every job.

Why it matters. See Risk 1. If this is wrong, the business cannot bill.

Feature items

6.1 — Three-dimensional rate lookup (*Account* → *Rate Card*) × *Freight Item* × *Zone*, resolved at a point in time.

Rate cards in production: *Clarendon and Domaine* (shared by two accounts), *Default Customer Rates*, *Tier 1–4*, *Test Customer*, *Wisdom Properties Group*. Resolution order: **named card** → **tier** → **default**.

6.2 — Effective-dated rate schedules Rates are versioned with start and end dates; pricing a job always uses the rates in force on that job's date.

Their current schedule runs 1 Apr 2026 → 1 Apr 2051. Non-negotiable: reissuing or crediting a March invoice must use March's rates.

6.3 — Two billing methods

- **Variable:** *service charge* (by zone) + ($\text{m}^2 \times \text{rate per m}^2$) + (*bags* × \$30)
- **Fixed price:** a flat amount per job, still varying by zone

Verified rates: Sydney \$220 + \$0.16/m² · Wollongong \$250 + \$0.18/m² · Newcastle \$250 + \$0.20/m². **Worked example:** a 1,000 m² Wollongong job with 2 bags = \$250 + (1000 × \$0.18) + (2 × \$30) = \$490 + GST.

6.4 — Per-customer component overrides Matt's exact requirement: *"Different customers may have different rates for the service charge or the weight charge component, and some customers have a fixed-price weight charge instead of a variable one."* Each component is independently overridable.

6.5 — Additional services — all nine, fixed and percentage

Charge	Rate	Type
Contamination Charge	\$90.00	fixed
Extra Load Time	\$100.00	fixed × qty (fractional — e.g. 3.65)
Fuel Levy	—	fixed
Fuel Levy – Wisdom	\$20.00	fixed, customer-specific variant
Fuel Levy (10%)	10%	percentage
Futile Pickup	\$120.00	fixed
Out of Area	\$2.00	fixed × qty
Recycling Bags	\$30.00	fixed × qty
Tipping Fuel Levy (7.5%)	7.5%	percentage

6.6 — Charge provenance and approval state Every charge records whether it was raised by a **driver** (needs approval), by the **system** (auto-generated), or by the **office** (manual), and carries its approval state and evidence.

6.7 — Auto-generated Extra Load Time ⚠ Time-on-site beyond an allowance, computed from driver arrive/depart timestamps.

Rule currently unknown — see Risk 2. If not answered by Day 3, ships as manual entry and is explicitly flagged.

6.8 — Job margin view Income vs cost per job, reproducing the margin figure Matt reads today.

Their current model applies a flat **\$100 "Reporting Cost (Pickup Rate)"** per job and displays e.g. *"Nett Total (Profit Margin = 83.28%)"*. We reproduce this, and can improve it later with real per-vehicle cost.

6.9 — Price check / preview tool Test a hypothetical job against any rate card without creating it — TransVirtual has this and the office uses it for quoting.

6.10 — Pricing regression harness (*internal, but a deliverable*) The Day 3–4 de-risking artefact from Risk 1, kept permanently in CI.

Workflows covered: underpins W8, W47, and all of M7

M7 · Invoicing & Finance

What it is. Turning priced jobs into invoices, getting them to customers, and getting them into the accounting system.

Who uses it. Office staff, Matt.

Why it matters. ~140 invoices/month, ~\$470 each, **7-day payment terms**. This is the cash cycle.

Feature items

7.1 — Invoice generation — F8 One invoice per job, generated on completion, numbered continuing from ~104,100.

7.2 — The two-invoice workflow — F10-adjacent, Matt's explicit requirement Base invoice issued **immediately** on completion against the original PO. Additional charges invoiced **separately, later**, once a new PO is received.

Matt's own words: *"any additional charges require a separate PO, which can take time. So, to avoid delays in receiving payment for the actual job, we send the invoice for the job immediately upon completion and then send another invoice later for any additional charges once a PO is received."*

Example: Job completes with a \$90 contamination charge. Invoice A (\$421.60) goes out that afternoon against PO [29916613/096](#). The \$90 moves to the **Awaiting PO** queue. Eleven days later Clarendon issues PO [29917104/012](#); office enters it; Invoice B (\$99 inc GST) is issued. Cash for the main job was never delayed. **Per-account rule:** accounts that don't require POs get everything on one invoice.

7.3 — "Approved charges awaiting PO" queue The workflow that doesn't exist today, and where money currently leaks.

Ageing visible, chase reminders, nothing forgotten.

7.4 — Invoice template resolution — Scope Call 1 The right template selected automatically by **type + priority + data filter**, exactly as TransVirtual does it.

Example: a Clarendon job resolves to *PlastaGo Recycling Invoice (kg & m2)* because Clarendon captures both; an iPlasta job resolves to *PlastaGo Recycling Invoice (m2 only)*.

7.5 — Six shipped templates + branding control PlastaGo m² · PlastaGo kg & m² · PlastaGo m²-only · EasyLift m² · EasyLift kg · RCTI layout. Self-service control of logo, colours, company details, payment terms, bank details, footer and visible columns.

All existing invoice content reproduced: ABN line, GBCA "Member 2026-2027" badge, direct deposit details (BSB 082-343, Acct 45 327 0863), 7-day terms, photo link, Meterage (m²) and Weight (kg) columns, customer reference / PO.

NEW **A functional GAIN over TransVirtual, not just parity.** Matt: *"this system won't allow me to put quantities in here ... in a perfect world I'd have **2 residential recycling bags at \$30 each, equalling \$60**, and then the weight charge at a rate of 16 cents per square metre."* [\[transcript 18:08\]](#) → Invoice lines must render **quantity × rate = amount**. TransVirtual cannot. Worth demoing.

Line	Qty	Rate	Amount
Service fee — Sydney	1	\$220.00	\$220.00
Residential Recycling Bag	2	\$30.00	\$60.00
Weight charge (<i>per m²</i>)	1,000	\$0.16	\$160.00

7.6 — PDF output — F49 PDF individually and in bulk. **CSV is optional, not required.**

↓ **Simplified by Call 2.** *"Really, we do **PDF the majority of the time**... [CSV] is really rare, I've only come across that once, and we don't even service that customer any more."* [C2 10:17] Build PDF properly; add CSV only if asked.

7.7 — Invoice list & bulk actions Search, filter by payment status and state, bulk approve, bulk send.

7.8 — Xero integration — F39, I1 **Push:** invoices and contacts to Xero. **Pull:** payment status back onto the invoice.

Scoped honestly as "2-way": invoices out, payments in, contacts synced. **Credit notes, part-payments and bank reconciliation are v1.1. Immediate benefit:** their current invoice list shows payment status *"Unknown"* for several customers — the sync is only partly working today. Fixing that is a visible day-one win.

7.9 — Customer payment history — F39, W53 Per-account view of what's been invoiced, what's paid, what's overdue.

Workflows covered: W53, W72, W73, and the finance side of W8

M8 · Notifications & Communications

What it is. The engine that proactively tells people things instead of waiting to be asked.

Who uses it. Everyone — customers, drivers, office.

Why it matters. Matt will judge this system by **whether it reduces phone calls**. That is the felt pain, even though the exception-evidence loop (M4) is the bigger financial win. Build both; **demo notifications first**.

Feature items

8.1 — SMS notifications — F12, I2 (Twilio) Job booked, allocated, driver on the way, completed, futile, rescheduled.

Example: *"PlastaGo: your pickup at Lot 328 Horologium Rd is scheduled for tomorrow (Wed 19 Aug). Reply or log in to reschedule."*

8.2 — Email notifications with photos — F24 Completion confirmations including the job photos.

— **F27 (separate post-job follow-up emails) is now OUT of scope.** The completion email with photos already covers the customer touchpoint; a second follow-up added little. **This is a promise they already make** — their booking form says *"Job alerts and photos will be sent to this email address"* (Ø3– §3). Today it's a URL on an invoice. We deliver it properly.

8.3 — Upcoming job reminders + 2-way reschedule — F17 + F65, W14 Automated reminder ahead of a scheduled pickup, prompting the site to confirm readiness — **with a link that lets them reschedule in one tap if it isn't ready.**

The problem, in Matt's words [transcript 33:51–35:57]: *"We get to some sites [and] the jobs aren't ready ... Letting the site supervisor know — hey, remember your job's booked in for this day, is it*

ready? — would be a really good step to have, to avoid us going to site."

His stated rules: if they don't reply, **go anyway**. If they reply not-ready, they give a new date and the job reschedules.

💡 **The refinement worth proposing.** Matt framed this as *replying to an SMS* (F65, marked v2 — inbound SMS parsing is fiddly and ambiguous). **A tap-through link is cheaper and better:**

"Job 61402 is booked for Wed 19 Aug. **Not ready? Pick a new date** →"

Opens the portal at that job's reschedule screen. No SMS parsing, no ambiguity, fully audited, identical for email and SMS.

✅ **F65 is now formally in scope** (the client's own list marks it "*nice-to-have v2*", so treat the tap-through link as the committed implementation and true SMS-reply parsing as the stretch). **Matt's stated rules:** if they don't reply, **go anyway**; if they reply not-ready, they supply a new date and the job reschedules [C2 33:51]. **This directly attacks futile pickups — the most expensive failure mode in the business.**

8.4 — Per-contact notification preferences Multiple contacts per account with roles (site / accounts / sustainability) and per-role channel preferences.

Why: today there is exactly one email — the site contact. The AP person who needs the invoice and the sustainability manager who needs the diversion data get nothing.

8.5 — Driver push notifications — F45 Covered in M4.11.

8.6 — Office ↔ driver communication — W50, W102 **Job-scoped comment threads** plus push notification, rather than a general chat product.

Scoped honestly: full real-time chat (F20) is v1.1. With two drivers, comments on the job record plus a push notification covers the need and keeps the conversation attached to the job it's about — which is better than chat for auditability. ✅ **Matt confirmed the shape:** "*It's between the **office and the driver***" — no driver-to-driver, no group channel. Office side = admin, allocator or office worker [transcript 37:30–38:44].

8.7 — Internal alerting on queues Daily digest and dashboard badges for the futile review, approvals and awaiting-PO queues.

This is the fix for the year-old unactioned futile pickup. The system must chase.

Workflows covered: W5, W14, W50, W58, W102

M9 · Reporting, Dashboards & Certificates

What it is. Fixed, well-designed reports and dashboards — **not** a custom report builder.

Who uses it. Matt, office staff, customers (self-serve).

Why it matters. Answers the five questions the owner currently cannot answer (05–\$4).

Feature items

9.1 — Monthly pickup volume report per customer — F1, W4 Jobs, m², weight, charges, by month, by customer, by site.

Parity requirement. They produce this today (**PlastaGo - Monthly Pickup Volumes**, AIO, and three versions of **Pickup Volumes**) and send it to customers. It must exist on day 20.

9.2 — Daily pickup volume trends — F2 Dashboard chart of jobs and m² per day.

9.3 — Monthly volumes by zone — F3 Sydney / Newcastle / Wollongong split.

Genuinely useful to them: zone mix drives margin, because the rate and the drive distance both vary by zone.

9.4 — Operations dashboard — F15 Jobs by status, jobs at risk of breaching target date, futile rate, contamination rate, completion times, unallocated jobs, queue counts.

Designed to answer, on one screen: "How many pickups are outstanding and which are about to breach?"

9.5 — Diversion Certificate — F52, W15, W59, W84 Branded PDF certifying material diverted from landfill — **issued per job**, and also available for a site, project or period. Generated automatically and emailed.

✓ **Matt named it and specified it** [C2 18:03]: "Some customers reporting to **Green Building Council** or **NABERS** need a certificate for their site stating how much waste has actually been diverted... **I'd call it a diversion certificate.**" **Content, his words:** "Your job of X amount of **square metres** had X amount of **waste**, and that has been successfully diverted from landfill." → **both m² and tonnage.** **Cadence:** "at the end of every job, a customer gets some sort of certificate", with the monthly report as supporting evidence. NEW **Blocked on a sample** — Decoded asked for one on the call so the data points are right. **Its headline number comes from M4.4.** The tonnes-diverted figure on the certificate is the **recovered weight**, reconciled against the tip-off weighbridge total — not the m² figure used for pricing. This is the strongest reason the tip-off reconciliation must be right: the certificate goes into a builder's Green Star submission and has to be defensible under audit (07- \$5A). **The single highest-value differentiator in the build.** Their customers need auditable diversion evidence for **Green Star**, council **Waste Management Plans** and their own ESG reporting (04- \$2.10). PlastaGo is a **GBCA member** and prints the badge on invoices, but has **no mechanism to issue the evidence.** Neither does TransVirtual. Neither does REGYP. **Example:** Clarendon's sustainability manager logs in and downloads "Clarendon Homes — 1 Jul 2025 to 30 Jun 2026 — 412 pickups, 387,400 m², 168 tonnes diverted from landfill" as a signed, branded PDF. That document is worth more to them than the recycling, and it is a switching cost no competitor currently matches.

9.6 — Financial summary ↑ **RESTORED to scope** Revenue, additional services, margin by customer and by zone.

Previously traded out to pay for enrolment. **The 20-day budget absorbs it.**

9.7 — Vehicle register & maintenance NEW — **F43** Per vehicle: **odometer + expense log** → auto-calculated **cost per kilometre**. Plus a **registration date with a 2-week reminder** that rolls forward on renewal.

✓ **Matt's exact spec** [C2 01:10]: "At 49,000 km it went in for a service, what the service description was — A service, B service — and the price. What it ends up doing is calculating... **a cost per kilometre.**" And explicitly **not** wanted: "I don't need all parts and labour and that breakdown. Keep it simple." **Rego:**

notify 2 weeks out; on renewal, log it and the date rolls forward by the registered period (12 / 6 / 3 months — configurable). **Why it matters now: 4 of their 5 vehicles currently show service dates in the past** (06- \$2). They have this module in TransVirtual and don't maintain it — so **the reminder is the feature**, not the register. Driver-reported defects (M4.9) land here.

9.8 — Driver profiles, licences & training NEW — **F53** Per driver: **licence photo, white card, crane tickets** (class 3 / class 4), each with an **expiry date** and an automated reminder.

✓ **Matt's spec** [C2 19:42]: reminder **1 month out**, and he preferred an **admin-managed licence-type register** — define the types once, set the lead time per type, apply against driver profiles. "Yeah, something like that... I think that would be better." **Compliance value:** Chain of Responsibility under the HVNL makes licence currency an operator obligation, not just the driver's (04- \$2.10).

9.9 — Driver performance metrics NEW — **F22** Jobs per day, on-site duration, futile rate, contamination rate, photo compliance, distance travelled, SLA adherence — per driver, over time.

Now feasible because F11 is full (M4.2) — continuous location gives real distance and drive-time data rather than estimates. ⚠️ **Two drivers.** Frame this as *operational insight and costing input*, not performance management — and say so when demoing it. Its real value is feeding the cost-per-km and job-duration models, which is where the money is.

Workflows covered: W4, W15, W59, W74, W76, W78, W82 — plus the *Data Analyst* persona, whose needs are met by these fixed reports in v1 (a dedicated analytics workspace is v1.1)

M10 · Data Migration & Cutover

What it is. Getting three years of TransVirtual data into the new system and switching over without breaking the business. **A module, not an afterthought.**

Why it matters. It is Risk 3 and Risk 4, and it is where fixed-deadline projects usually die. **The extra six days exist partly to buy this room** — migration gets days 17–18, not an afternoon.

Work items

10.1 — Prove the export on Day 1 Before any build decision is locked, confirm TransVirtual can bulk-export consignments, invoices, **photos**, accounts, rate cards and additional services.

10.2 — Migration inventory

Object	Volume	Notes
Consignments (jobs)	5,079	38 months
Accounts	53 total	31 active in last 12 months; top 5 = 87% of jobs
Invoices	thousands	
Photos	~40–50k est.	⚠️ Biggest risk. ~8–10 per job × 5,079 ≈ 80–100 GB. Prove bulk export Day 1

Object	Volume	Notes
Rate cards	8	+ effective-dated schedules
Freight items	19 (7 current)	Includes hook bins
Additional services	9	
Vehicles	5	
Users / drivers	2 active, ~7 historical	
Templates	16	
Mobile forms	1	Driver Pre Start Checklist

10.3 — Resolve the m²/kg convention (Risk 3) before transforming anything.

10.4 — Timezone normalisation — NZST/NZDT → Australia/Sydney.

10.5 — Sequence continuity — jobs continue from ~61,300; invoices from ~104,100.

10.6 — Parallel-run validation — the pricing harness, plus a reconciliation of migrated totals against TransVirtual reports.

10.7 — Cutover runbook and rollback plan — including the two-week read-only overlap (Scope Call 3).

10.8 — UAT with Matt and both drivers — Days 13–14. The drivers must physically use the app on a real run before go-live.

6A. Technical architecture & stack

Status: APPROVED. Final stack as agreed. Where a decision was contested during selection, the reasoning is recorded so nobody re-opens it later.

6A.1 The stack

Layer	Decision
Language	TypeScript across web and API. Non-negotiable for AI-assisted codegen — types are what keep generated code coherent across ~40 screens and two client apps
Database	MongoDB Atlas — ap-southeast-2 (Sydney). ⚠️ Atlas, not DocumentDB — transactions and change streams need a genuine replica set and full Mongo compatibility
ODM	Mongoose + TypeScript (Typegoose optional). ⚠️ Not Prisma — its Mongo connector is weakest exactly where this build needs strength: aggregation pipelines and multi-document transactions
API	Express + TypeScript — one service, plus BullMQ worker processes
Web	React + Vite + TypeScript

Layer	Decision
Routing	React Router (<i>TanStack Router if type-safe routes are wanted</i>)
UI kit	Tailwind + shadcn/ui — ★ the single biggest lever on the Day-3 target. Components are copied in and owned, not a black-box dependency, and they generate very reliably
Grids	TanStack Table — the client's entire mental model is grids; they are coming from jqGrid
Data fetching	TanStack Query
Forms + validation	React Hook Form + Zod — Zod schemas shared between web and API
Charts	Recharts
Driver app	★ BOTH, in full — React PWA (vite-plugin-pwa + Workbox) and Flutter
Queues	BullMQ + Redis
Object storage	S3 ap-southeast-2 (Sydney) — presigned direct uploads from the driver apps
PDF generation	Playwright (HTML→PDF) + pdf-lib for merging — see §6A.6
Auth	OTP → short-lived JWT + refresh token. httpOnly cookie on web; Keychain/Keystore on Flutter
Email / SMS	M365 SMTP (I5) · Twilio (I2)
Hosting	Render (Singapore) — see §6A.2
CI/CD	GitHub Actions — ★ the pricing regression harness runs here
Error tracking	— Removed by decision — see §6A.8

6A.2 Hosting and data residency — the one thing to disclose

Render has no Australian region. Its regions are Oregon, Ohio, Virginia, Frankfurt and Singapore; the closest to Sydney is **Singapore**. That matters because the client's Privacy Policy commits to the Australian Privacy Principles and a 7-year retention floor (02- §6).

The agreed split keeps all persistent data onshore:

Component	Location	Onshore?
MongoDB Atlas — all business data	ap-southeast-2 Sydney	✓
S3 — photos, PDFs (~80–100 GB)	ap-southeast-2 Sydney	✓
Express API + BullMQ workers	Render Singapore	✗ compute offshore
Redis — queue payloads (<i>transient</i>)	Render Singapore	✗ transient only
Static web + driver PWA	Render CDN	—

Latency Sydney↔Singapore is ~90–100 ms — irrelevant at 7 jobs/day.

⚠️ **Action, not a blocker:** their **Privacy Policy should disclose overseas processing**. It is a one-paragraph amendment under APP 8. Raise it with Matt proactively — far better than a builder's procurement team finding it during vendor due diligence.

If fully-onshore compute is ever required: Fly.io and AWS (App Runner / ECS Fargate) both have Sydney regions. The application is portable; only the deploy target changes.

6A.3 ⚠️ Eight things that must be true for MongoDB to be safe

MongoDB was chosen over PostgreSQL. That is workable at this scale — transactions, **Decimal128**, schema validators and the aggregation pipeline all exist. **But the database will not enforce integrity for you, so these eight items are requirements, not suggestions.**

#	Requirement	Why
1	Decimal128 for every monetary field — rates, charges, GST, totals. Never double	The single highest-consequence line in the schema. GST and rounding must match TransVirtual to the cent (Risk 1)
2	Atlas, not DocumentDB	Transactions and change streams need a real replica set and full compatibility
3	withTransaction sessions around invoice generation	Job → charges → invoice → sequence increment must all commit or none do
4	Collection-level \$jsonSchema validators , not only Mongoose schemas	Mongoose validates only writes that go <i>through</i> Mongoose. The DB validator catches everything else — belt and braces on financial collections
5	One data-access layer; no ad-hoc queries	Referential integrity now lives in application code, so it must live in one place — repository modules per aggregate, and nothing else touches the driver
6	Compound indexes designed up front	Two hot paths: effective-dated rate lookup (rateCardId, freightItemId, zoneId, effectiveFrom) and reporting (accountId, completedAt) / (zone, completedAt)
7	Change Streams → append-only audit collection	Satisfies M1.6. Genuinely nicer in Mongo than in SQL — take the win
8	\$merge rollup collections for monthly volumes by customer and zone, recomputed nightly	Those reports are read constantly; don't re-aggregate thousands of jobs on every request

The structural consequence

In a relational database, constraints would catch a whole class of integrity bug for free. Here **the test suite is doing that job**. So the **pricing regression harness stops being a nice-to-have and becomes structural** — Risk 1 gets *more* important, not less, and the harness must run on every commit from the moment the pricing engine exists.

6A.4 Two full driver apps — cost and mitigation

Both the React PWA and the Flutter app ship complete. Honest sizing:

	Effort	Owner
Driver PWA	~3–4 days	Web devs. UI is nearly free after the Day-3 design lock — but the offline queue (IndexedDB/Dexie + service worker), photo queue and sync conflict handling are not
Driver Flutter	~10–12 days	Mobile dev (has 20)
Shared sync protocol	~1 day, once	Specified by one person, implemented twice

Flutter fits comfortably. The PWA is the pressure point — it lands on the web devs on top of the admin console, customer portal and enrolment.

Two mitigations, both mandatory:

1. **Specify the offline sync protocol once, in writing, before either implementation starts.** Operation log · idempotency keys · last-write-wins with server arbitration · explicit conflict cases · photo upload as a separate resumable queue. *Two implementations against one written contract is safe. Two implementations against a vague shared understanding is how they silently diverge.*
2. **Keep a feature-parity checklist** as a living test document. Every driver feature is now a two-place change, and the second place is the one that gets forgotten.

What the PWA buys in return: the day-one fallback while Apple enrolment clears (Risk 8), and a driver whose phone dies can work from any browser.

6A.5 Repo shape

```
plastago-web/                                ← npm workspaces + Turborepo
├─ apps/
│  ├─ web/      admin console + customer portal  (one Vite app, role-based
routing)
│  ├─ driver/   driver PWA                        (separate Vite app – service
worker, offline shell)
│  └─ api/      Express + BullMQ workers
└─ packages/
   ├─ shared/   Zod schemas → single source of truth → generated OpenAPI
   └─ ui/       shadcn components shared by web + driver

plastago-mobile/                             ← SEPARATE repo, Flutter
```

Why two Vite apps and not three: admin and portal share auth, session and most components, so one app with role-based routing is simpler. The driver PWA is separate because the service worker and offline shell need their own build configuration.

⚠️ **The consequence of a separate mobile repo:** the **OpenAPI spec becomes a published artefact**, not a shared file. Have the web repo's CI emit a **versioned spec** (and optionally a generated Dart client package) so the Flutter repo **pins a version** rather than chasing a moving target. Without this, two repos drift silently — it is the main risk of splitting them.

6A.6 One stack choice that changes scope: HTML-based PDFs

Invoices, Diversion Certificates and the SRA+SWMS pack are generated as **HTML+CSS rendered to PDF** (Playwright), with **pdf-lib** merging the static SWMS page.

★ **This directly de-risks Scope Call 1.** Matt wants to edit invoice layouts, and his current tool is a binary banded report designer. Because our templates are **HTML+CSS**, a **constrained editor** — logo, colours, company details, terms, bank details, footer, visible columns, free text blocks — becomes a genuinely small feature instead of rebuilding Stimulsoft. **Choose this format for that reason, not only for rendering fidelity.**

6A.7 Express needs structure imposed on day 1

Express provides none, and AI-generated code drifts without a convention. Decide it once, write it in the README, and generated code stays coherent:

- **router → controller → service → repository**, one folder per domain
- **Zod validation as middleware** on every route, sourced from **packages/shared**
- **helmet**, rate limiting, CORS allowlist
- A **central async error wrapper** — no unhandled promise rejections
- **Repository layer is the only code that touches Mongoose** (§6A.3 #5)

6A.8 No error tracking — the accepted gap and its mitigation

Error tracking was **removed by decision**. The honest consequence: **production failures are invisible unless a human reports them** — and with two offline-first driver apps, the realistic failure is a **silent sync failure** that nobody notices for days.

Mitigation without adding a vendor:

- **Structured JSON logs** from the API and workers, retained on Render
- **A sync-failure counter and last-successful-sync timestamp per driver device, surfaced on the admin dashboard (M9.4)** — so a stuck queue is visible in the product, not only in logs
- **BullMQ failed-job counts on the same dashboard** — dead-letter jobs must be seen
- Health endpoint + Render alerting on service restarts

This is a deliberate trade, recorded so it is a known gap rather than an oversight. Revisit if silent sync failures actually occur.

6A.9 The API contract is a Day-3 deliverable

Because the frontend is built first against mock data (§13), **the API contract falls out of real UI needs rather than being guessed**. Treat it as a first-class artefact:

- **Zod schemas in packages/shared** — the single source of truth
- **Generate OpenAPI** from those schemas

- **Flutter codegens its client from the OpenAPI spec** — this is the integration surface between the two ecosystems, and it is what makes a separate mobile codebase safe
- Mock with **MSW** during days 1–3 so the frontend is genuinely runnable at the demo
- **Freeze the contract at day 3**. Changes after that need a version bump, because two consumers depend on it

6A.10 Non-negotiables carried from the analysis

1. **Money as Decimal128**. Never floats. GST and rounding must match TransVirtual to the cent.
2. **area_m2 and weight_kg as first-class typed fields** — never overloaded (Risk 3, M1.3).
3. **Effective-dated rate documents** with point-in-time resolution (M6.2).
4. **Append-only audit collection** fed by Change Streams (M1.6).
5. **Store UTC, render Australia/Sydney**; migration converts from NZST/NZDT (M10.4).
6. **7-year retention; all persistent data in Sydney** (§6A.2).
7. **Job and invoice sequences continue** from ~61,300 and ~104,100 (M1.4) — needs a transactional counter document, not a naive \$inc race.
8. **Multi-brand as a first-class dimension**, not a config flag (M1.1).
9. **Presigned direct-to-S3 photo uploads** — never stream photos through the API.
10. **The pricing regression harness lives in CI and gates deploys** (Risk 1, §6A.3).

7. Scope summary — Features

✓ IN — the 35 agreed features

ID	Feature	Module
F1	Monthly pickup volume reports per customer	M9.1
F2	Daily pickup volume trends	M9.2
F3	Monthly pickup volumes by zone	M9.3
F4	Record and display m ² and weight per job	M1.3, M2.3, M4.3
F5	Itemised job charges	M2.3, M6, M7.5 (qty × rate × amount)
F6	Unique job numbers	M1.4, M2.2
F7	Link job photos to job records	M2.3, M4.5
F8	Generate invoices for jobs	M7.1
F9	Customer portal	M5 Part 1
F11	GPS tracking — FULL, real-time driver location ↑	M4.2 (<i>native shell</i>)
F12	Customer SMS notifications	M8.1
F14	Driver photo upload · Site Risk Assessment · SWMS PDF	M4.5, M4.8
F15	Admin dashboard — job completion times	M9.4

ID	Feature	Module
F17	Automated reminders about upcoming jobs	M8.3
F18	Customers view job history online	M5.8
F21	"Mobile app" for customers → PWA portal (\$2.1)	M5 Part 1
F22	Driver performance metrics NEW	M9.9
F24	Real-time job completion notifications	M8.2
F26	Customer dashboard — real-time job tracking	M5.7
F28	Request urgent pickups / modify urgency	M5.5
F29	Manage customer accounts and preferences	M2.8, M5.15
F30	Track job delays and reasons	M2.5
F33	Driver schedules via Run Sheet	M3.2
F34	Weight entry during pickups	M4.3, M4.4
F38	Job cancellation and rescheduling	M2.4, M5.4
F39	Customer payment history — Xero 2-way	M7.8, M7.9
F43	Vehicle maintenance + cost/km + 2-week rego reminder NEW	M9.7, M4.9
F45	Push notifications to drivers	M4.11
F46	Customers set preferred pickup times ↑ restored	M5.6
F49	Invoices in PDF (+ CSV)	M7.6
F52	Diversion Certificate (PDF, emailed)	M9.5
F53	Driver training / licence records + expiry reminders NEW	M9.8
F54	Track job pickup delays and notifications	M2.5, M3.5
F56	Site-specific hazards / Site Risk Assessment	M4.8
F65	2-way reschedule response for upcoming jobs NEW	M8.3

Plus, not on the client's list but required for parity or correctness: multi-brand support · party model · rate card engine · **tip-off weight capture & deduct-and-average reconciliation (M4.4)** · futile review queue · additional service approvals queue · awaiting-PO queue · two-invoice workflow · invoice template assignment · **qty × rate invoice lines** (a gain over TransVirtual) · job margin view · price check tool · site/project register · audit log · geocoding · **customer enrolment (M5 Part 2)** · data migration.

⚠ Customer enrolment (M5 Part 2) has no F-number but is non-negotiable. Without it there is no intake on day 21 — the current booking form is public and the new portal is authenticated. See Risk 6.

— Dropped from the earlier 14-day scope

ID	Feature	Why it went
F27	Post-job follow-up emails	The completion email with photos (F24) already covers the touchpoint
F41	Driver job-completion feedback flagged for review	A plain note field remains; Extra Load Time is system-generated anyway, so no commercial impact
F32	Customers submit job-related documents	Customers email the office as today; small add in v1.1

🚫 OUT — see §11 for reasons and versions

F10 (RCTI) · F13 · F19 · F20 · F27 · F31 · F32 · F35 · F40 · F41 · F55 · F57 · F58

8. Scope summary — Integrations

Six agreed IN. Two dropped.

ID	Integration	Verdict	Scope
I1	Xero	✅ IN	Push invoices + contacts; pull payment status. Credit notes, part-payments and bank rec → v1.1. ✅ Confirmed live (not MYOB): <i>"I'm not looking to replace Xero, I'm looking to interface with it"</i>
I2	Twilio	✅ IN	SMS notifications + SMS OTP authentication
I3	Google Maps Platform	✅ IN	Address autocomplete, geocoding, map pins, live driver map (now that F11 is full), driver navigation hand-off. Not route optimisation (F19 → v1.1)
I5	Microsoft 365	✅ IN	Outbound email via M365 SMTP — replaces SendGrid at Matt's request. ⚠️ Caveat to raise: M365 SMTP is throttled (~30 msg/min, ~10k recipients/day) with weaker bounce handling than a dedicated provider. Fine at ~140 jobs/month, but a conscious choice
I6	Microsoft 365 Outlook	✅ IN ⬆️	Inbound mailbox monitoring for the PO pipeline (M2.12). Moved in from v1.1
I8	Mistral AI Document AI OCR	✅ IN ⬆️	PO extraction from email (M2.12). Moved in from v1.1 — this is the client's stated core pain. ⚠️ Risk 9 — must ship with confidence thresholds and a human review queue
I4	Stripe	🚫 DROPPED by the client	Matt: <i>"a lot of our payments come through direct deposit... I'm probably happy to take that requirement out."</i> Payment is 7-day EFT against PO

ID	Integration	Verdict	Scope
t7	Cognito Forms	DROPPED — do not integrate	We are replacing it. Two intake paths = two sources of truth. /book-a-pickup stays as a public branching page (Scope Call 2 / M5 · A.1) with the dead iframe removed

Also present but not on the list: a **MYOB** account link exists in TransVirtual alongside Xero. Confirm dead, then drop.

A third system surfaced on Call 2: a separate CRM for proposals (Matt: "a separate CRM that's f***ing terrible"). Portal-based proposals are **v2** — but identify the tool now, since it may hold lead data and is a third subscription to retire. See Q34.

9. Scope summary — Authentication

ID	Method	Verdict	Notes
A1	Email OTP	IN	Primary for office, admin and customer administrators
A2	Phone / SMS OTP	IN	Primary for site supervisors and drivers — the right choice for people on building sites who will never remember a password. Uses the Twilio integration already in scope

Also in scope:

- **Single sign-on across all surfaces** — admin console, customer portal, **and both driver builds (native + PWA)**. This is what delivers Matt's "shared user credentials" requirement. **The native shell needs secure token storage** (Keychain / Keystore), not browser localStorage — a small but real addition for the mobile developer
- **Role-based access control** across the 7 roles (M1.5)
- **Single-tenant** architecture, per the client's spec
- **Session management**, device registration for drivers, and audit of all logins

10. Scope summary — User Workflows (all 109)

Note on the count. The brief numbers workflows up to **W122**, but the numbering has gaps — there are **109 actual items**. Of those, **78 are IN** and **31 are OUT**.

Net effect of the 20-day scope change: W103, W113, W115 come in with F22 and F43; W55, W58, W91 go out with F41, F27 and F32. **Total unchanged at 78.**

Read §2.3 first. ~80 of the 109 are permutations of ~30 real screens. The IN count is high because one module satisfies many workflows at once — not because we are building 78 separate things.

IN — 78 workflows (3 added: W103, W113, W115 · 3 removed: W55, W58, W91)

Administrator — 11 IN of 20

ID	Workflow	Module
W1	Global Administrator	M1.5
W2	Manage user accounts	M1.5
W3	Configure system settings	M1, M2.8
W4	Generate monthly reports	M9.1
W5	Manage system notifications	M8.4
W7	Manage system integrations	M7.8, M8.1
W8	Manage customer accounts	M2.8
W14	Automated reminders for customers	M8.3
W15	Generate certificates of recycling	M9.5
W16	Configure user permissions	M1.5
W17	Manage system backups	M1.7 (<i>automated infrastructure, not a UI</i>)

Driver — 16 IN of 19

ID	Workflow	Module
W25	Drivers (role)	M1.5, M4
W26	Log job completion	M4.2
W27	Report issues on site	M4.6, M4.7
W28	Take job photos	M4.5
W29	Update job status	M4.2
W30	Log pickup weights	M4.3
W31	Review job history	M4.1
W32	Provide feedback on job sites	M4.7 (<i>contamination assessment; F41 review flow dropped</i>)
W33	Log maintenance issues on vehicles	M4.9 → M9.7 (<i>now the full F43 feature</i>)
W34	Upload job site photos	M4.5
W35	View upcoming job schedule	M4.1
W36	Receive real-time job updates	M4.11
W37	Log site-specific hazards	M4.8
W38	View real-time job status	M4.1
W41	Log job-specific hazards	M4.8
W44	Log recycling bag weight	M4.3

Office Worker — 16 IN of 23

ID	Workflow	Module
W45	Office workers (role)	M1.5
W46	Process purchase orders	M2.10 (<i>manual</i>) + M2.12  automated AI extraction from email
W47	Generate job cards	M2.1
W48	Attach POs to job cards	M2.10
W49	Manage job documentation	M2.11
W50	Coordinate with drivers	M8.6
W51	Review job requests	M2.2
W52	Coordinate with customers	M8, M2.11
W53	Track customer payment history	M7.9
W55	Track job completion feedback	— OUT — F41 dropped. A plain note field remains
W56	Manage job-related documentation	M2.11
W58	Automated follow-up emails post-job	— OUT — F27 dropped. Completion email with photos covers it
W59	Generate certificates of recycling	M9.5
W61	Manage customer job request history	M2.2
W62	Manage job cancellation requests	M2.4
W65	Track job delays and reasons	M2.5
W67	Job cancellation and rescheduling	M2.4
W68	Assist with customer inquiries	M2.11

Customer – Office Administrator — 14 IN of 15

ID	Workflow	Module
W70	Site Supervisor management	M5.14
W71	Update customer information	M5.15
W72	View customer invoices	M5.10
W73	Review invoices	M5.10
W74	Access monthly reports	M5.11

ID	Workflow	Module
W75	Manage customer job requests	M5.1, M5.7
W76	Review monthly reports	M5.11
W77	View historical job completion times	M5.8
W78	Request job completion reports	M5.11 (<i>self-serve</i>)
W80	Manage job request prioritisation	M5.5
W81	Manage account preferences	M5.15
W82	Request job-specific reports	M5.11 (<i>self-serve</i>)
W83	Manage customer job history	M5.8
W84	Request job-specific compliance reports	M9.5 (<i>recycling certificate; EPA reports → v1.1</i>)

Customer – Site Supervisor — 10 IN of 14

ID	Workflow	Module
W85	Site supervisors (role)	M1.5
W86	Confirm job readiness	M5.2
W87	Provide site access information	M5.3
W88	Submit job request	M5.1
W89	Request urgent pickups	M5.5
W91	Submit job-related documents securely	— OUT — F32 dropped. Email to the office as today
W92	Set preferred pickup times	M5.6
W95	Request job status updates	M5.7 (<i>becomes self-serve — no request needed</i>)
W97	Request job rescheduling	M5.4
W98	Provide job site access updates	M5.3
W99	Request job completion feedback	M5.9 (<i>self-serve completion record</i>)

Allocator / Driver Manager — 11 IN of 18

ID	Workflow	Module
W100	Allocate jobs to driver / create run sheets	M3.1, M3.2
W101	Review job assignments	M3.1
W102	Communicate with drivers	M8.6
W104	Reassign jobs	M3.1
W108	Track job pickup delays	M3.5

ID	Workflow	Module
W111	Track driver availability	M3.4 (<i>simplified column</i>)
W120	Track job delays and reasons	M2.5
W122	Job cancellation and rescheduling	M2.4
W103 NEW	Monitor driver performance	M9.9 (<i>F22 now in scope</i>)
W113 NEW	Manage and schedule vehicle maintenance	M9.7 (<i>F43 now in scope</i>)
W115 NEW	Track job-specific driver performance	M9.9 (<i>F22 now in scope</i>)

Plus (from the client's PDF but unnumbered in the list): *Allocator* · *Review tip-off weights* → **M4.4** (capture + reconciliation) · M9.5 (feeds diversion certificates) — **IN**

OUT — 31 workflows (+ *W55, W58, W91 dropped above*)

ID	Workflow	Version	Reason
W9	Configure reporting settings	v1.1	Fixed reports in v1; parameterisation later
W10	Manage subcontractor job assignments	✗ REMOVED	✅ Matt: "That can probably go, because it's a driver. They're all subcontractors " [C2 21:33]. Ordinary driver allocation already covers it — it was never a separate workflow
W11	Configure reporting templates	v1.1	Template <i>authoring</i> — Scope Call 1
W13	Implement user feedback	—	A business process, not a software feature
W18	Manage recycling centre capacity	v3	Processing side not in scope
W20	Configure automated job assignment rules	v1.1	Only 2 drivers — manual is better
W22	Configure customer support workflows	v1.1	Ticketing not in v1
W23	Manage quality assurance processes	v2	Undefined
W24	Analyse customer feedback trends	v1.1	Needs feedback capture (F13) first
W40	Receive route optimisation updates	v1.1	Needs routing engine (F19)
W42	Receive real-time weather updates	v2	Marked nice-to-have by client

ID	Workflow	Version	Reason
W43	Log fuel consumption	v2/v3	Marked v2/v3 by client
W54	Manage and track customer contracts	v2	Marked v2 by client
W63	Track job-specific compliance documents	v1.1	Marked v2 by client
W64	Track job-specific waste disposal locations	v3	Marked v3 by client
W66	Manage and track job-specific permits	v1.1	Marked v2 by client
W69	Manage customer feedback submissions	v1.1	Needs F13
W79	Manage customer feedback submissions	v1.1	Needs F13
W90	View real-time driver ETA	v1.1	Marked "very-nice-to-have v2"; needs routing
W94	Provide job site feedback	v1.1	Needs F13
W96	Provide feedback on driver performance	v1.1	Needs F13
W107	Optimise job allocation	v1.1	Needs routing engine
W109	Implement automatic job assignment	v1.1	Only 2 drivers
W110	Analyse job allocation efficiency	v1.1	Needs routing data
W117	Conduct site inspections	v2	Not a software workflow as stated
W118	Implement compliance training for staff	v2	Not a software workflow as stated
W119	Track job-specific waste disposal locations	v3	Duplicate of W64

ID	Workflow	Version	Reason
W121	Manage and track job-specific permits	v1.1	Duplicate of W66
—	Data Analyst persona (7 workflows in the PDF)	v1.1	Needs met by M9 fixed reports in v1; a dedicated analytics workspace with custom report building (F31) is v1.1

11. Out of scope — v1.1, v2 and drop

v1.1 — the immediate follow-on (weeks 5–8)

ID	Feature	Why not in the 20 days
F10	RCTI (Receiver Created Tax Invoices)	⚠️ My earlier reasoning was wrong. I said "zero active subcontractors" — in fact "basically all of our guys are subcontractors" [transcript 23:59]. It is still deferrable because Matt says <i>"we don't really use it at the moment, but we're planning to"</i> — but it is a near-term need, not a distant one. Mechanics now known: agreed per-job rate, jobs accumulate, weekly run, exports to Xero as a BILL (accounts payable), not an invoice. Driver pay rates should enter the data model early even if the RCTI run doesn't ship

| **F13** | Customer service ratings | Client marked nice-to-have. Unblocks W24, W69, W79, W94, W96 | | **F19** | **Route optimisation** | Client marked *"very-nice-to-have"*. **Geocoding + visual clustering ships in v1**, which is the prerequisite. 💡 **Reframe it for the v1.1 conversation:** the real driver isn't efficiency, it's **succession**. Matt: *"my allocator used to be a truck driver, he looks at a sheet of suburbs and he knows what to do ... [but] he easily is 50, so in the future he may not be available"* [transcript 36:44]. F19 is knowledge capture for a single point of failure | | **F20** | Office ↔ driver chat | Job comments + push notifications cover it for 2 drivers, and keep conversation attached to the job | | **F31** | Custom report builder | A product in itself. Fixed reports in v1 | | **F35** | **Recycling compliance tracking (EPA)** ↓ moved v1.1 → v3 | ⚠️ **Corrected by Call 2.** I had this as legally-exposed and near-term. It isn't: *"the yard that we've got, we don't operate it under an EPA licence at this time, we're more of a transfer station, but we are planning to do the manual processing on-site in the future"* [C2 08:19]. **They are not the processor**, so RRO14 does not bind them today — a third party processes and bills them monthly. This becomes real only when processing comes in-house. **Matt's own call: v3.** See 09- \$20 | | **F40** | Driver availability dashboard | 2 drivers. A column on the allocation board suffices |

| **F55** | Customer-visible estimated completion times | Needs the routing engine | | **F57** | Customer complaints tracking | Needs a ticketing model | | **F58** | Automatic job assignment → actually **proximity dispatch** | ⚠️ **I mis-scoped this.** It is not availability scheduling. Matt: *"if they're in a certain area and a job comes in... there's already a driver in that area"* [C2 23:40] — i.e. flag a late booking to a driver who is already nearby. More useful than I assumed, but it **depends on live driver location**, so it is gated on the Risk 7 / Q6 GPS decision | | — | **Invoice template WYSIWYG designer** | Scope Call 1 — the biggest single scope trap | | — | Xero credit notes, part-payments, bank reconciliation | | — | Bag asset tracking (QR/barcode, bag lifecycle) | ✅ **Matt scoped this to "version three" himself.** Real problem though: bags go free to Knauf in **pallets of 190**, ship with board orders, and *"we've found our bags on sites that we're not servicing"*. He wants *"we've sent out a thousand bags and we've only collected 900"* [transcript 49:11] |

v2 and beyond — **versions below are Matt's own calls from Call 2**

Item	Ver	Note
Proposals / quotes in the customer portal	v2	 A third system exists — a separate CRM he calls " <i>f***ing terrible</i> " because it forces per-pickup line items onto open-ended multi-site proposals. Wants proposals visible in the portal to download, accept or decline. Validates M5 Part 2 Journey A. 09- \$27
F44 — job-specific permits	v2	Client-supplied permits with time windows (traffic control near a school). " <i>Just another document assigned to that job.</i> " Decoded floated a v2 AI agent to extract the permit window and auto-schedule within it
Customer-specific report templates	v2	But the monthly pickup volume report is essential in v1 — " <i>we have one customer in particular that requires it</i> "
F65 2-way reschedule-by-reply	v1.1	Largely covered by the tap-through link in M8.3
Recycling centre capacity	v3	Needs an EPA licence they don't hold
Historical recycling rates per site	v3	" <i>That's when the data analysis features will come in</i> "
Waste disposal locations · recycling input/output	v3	Ties to processing they don't yet operate
Bag asset tracking	v3	Matt's call
Weather · per-job environmental impact · fuel consumption	v2	" <i>one of those AI suggestions</i> "
Processing batch / mass balance / yield · gypsum sales & stock	v3	Depends on in-house processing
Knauf/DPO channel integration	v2	

 Removed entirely — Matt struck these on Call 2

Item	Why
Job-specific recycling rates	" <i>We achieve 100% recycling rate anyway. If it comes into our facility, it gets recycled</i> "
Recycling bag usage per job	Redundant — " <i>the driver is going to say how many bags are on site anyway</i> "

Item	Why
Real-time job status updates (as a discrete feature)	Redundant — everything is push-driven and real-time by design
W10 subcontractor job assignments	Redundant — all drivers are subcontractors; normal allocation covers it

Drop / challenge

Item	Recommendation
I4 — Stripe	Drop. Client's own spec says " <i>possibly not needed – v2</i> ". Payment is 7-day EFT against PO; builders' AP departments do not pay by card
I7 — Cognito Forms	Drop, do not integrate. We are replacing it. Two intake paths = two sources of truth = the exact re-keying problem we're solving
MYOB link	Confirm dead, then drop
~80 of 109 workflows as separate backlog items	Treat as views of ~30 screens (\$2.3)

12. Question register

Complete and current. Grouped by *who answers and when*, not by topic. Every question raised anywhere in this knowledge base appears here; answered ones stay in **\$12.5** so nothing gets re-asked.

\$12.1	🔴	5	Blocking — ask Matt before Day 1
\$12.2	🔧	2	We answer these ourselves from the TransVirtual tenant
\$12.3	🟠	13	Needed during week 1
\$12.4	🟡	13	Useful, answerable during the build
\$12.5	✅	26	Already answered and closed

33 open questions of 37 raised. Numbering is stable — gaps (Q8, Q9, Q13, Q33) are deliberate: those are answered and live in \$12.5.

12.1 🔴 TIER 1 — BLOCKING. Ask Matt **before Day 1**

These change architecture, scope or the plan itself, and **only Matt can answer them**. Do not start without them. (*Five questions — Q2 and Q4 moved to \$12.2; Q8 removed.*)

#	Question	Why it blocks	Ref
Q1	Do you need 3 years of history migrated, or is a read-only archive acceptable?	The single biggest scope lever. "Archive is fine" removes several days from M10	Risk 3, M10

#	Question	Why it blocks	Ref
Q3	What is the m²/kg field convention — per era and per customer?	The meaning of both columns has drifted at least twice in 3 years. Blocks all migration transforms	Risk 3
Q5	Confirm the invoice-template split: resolution + branding now, WYSIWYG authoring v1.1	Unbounded scope if left open. Needs agreeing in writing	Risk 5, Scope Call 1
Q6	Do you need live background GPS (moving dot with the phone in a pocket), or is location captured at each status event enough?	Decides PWA vs native for the driver app.  ANSWERED — build both (native + PWA)	Risk 7, M4.2
Q7	 Who actually raises bookings today — head office, or site supervisors on site?	If Clarendon's contracts administrator raises all ~90/month centrally, Journey B mostly evaporates and M5 Part 2 simplifies a lot. If it's per-site supervisors, the site-link flow is the most important screen in the build	M5 Part 2

Q2, Q4 → moved to §12.2. Both are answerable from the TransVirtual tenant ourselves — no need to spend Matt's time on them.

Q8 (TransVirtual notice period) → removed. It is a commercial matter on the client's side, not a build blocker. Scope Call 3 only needs Matt's *agreement* to the two-week overlap, not the contract date.

12.2 ANSWER OURSELVES — portal investigation, don't ask Matt

We have read access to the production tenant. These get resolved by looking, not asking. **Findings below are already recorded** — what remains is stated explicitly.

#	Question	What the portal already tells us	What's left
Q2	Can TransVirtual bulk-export photos, and how many are there?	 Export mechanism confirmed working in production. Import/Export → Data Export → Photos exists and is rule-based. Zero image-export rules configured today — but four POD export rules are live and running: <i>PlastaGo Pickup Complete Photos, PlastaGo Futile Pickup Notification, EasyLift Pickup Complete Photos, EasyLift Futile Pickup Notification</i> . Those are what email photos to customers today, which proves the export pipeline works	Create an image-export rule against a date range and measure actual volume + throughput on Day 1 . Estimate stands at ~40–50k images / 80–100 GB
Q4	"Extra Load Time" — free allowance and unit?	 Identified: it is TransVirtual's demurrage (wait time) feature. Global Setup tooltip: "...confirm if they are on site. If they are, arrival time is locked in place and the system will start to calculate possible demurrage (wait time) "	 The free-time threshold binding is not in Global Setup. Likely per

#	Question	What the portal already tells us	What's left
		charges, if configured. Ask Driver to Confirm Onsite = ON , and ClientMobileWarnDriverOnsite15min = ON (a 15-minute on-site warning). Charge type config: flat rate \$100, requires approval, single-use per consignment, and NOT driver-raisable (Display to driver unchecked) — which is why the queue shows <i>Created By: System</i>	rate card or per customer. Next step: check the rate card's "additional service price override" screen (visible in Matt's emailed screenshots) before falling back to asking him

12.3 ● TIER 2 — Needed during **week 1**

Each of these blocks a specific module once we reach it.

#	Question	Why it matters	Ref
Q9	Where do you tip off, and whose facility is it?	✅ ANSWERED (Call 2). A yard run as a transfer station, no EPA licence, processing outsourced. Only <i>who</i> the processor is remains open — see Q34	09-\$20
Q10	Is ANY current customer billed on kg? PlastaGo Recycling Invoice (KG) and EasyLift Recycling Invoice (KG) templates exist	If yes, tip-off accuracy becomes a billing input, not just reporting — which raises its risk class considerably	07-\$5A.3
Q11	EasyLift — which customers? Did it bill on kg while PlastaGo bills m²?	⚠️ Partly answered from the portal: EasyLift is OPERATIONALLY LIVE, not legacy. It has two active POD export rules running in production (<i>EasyLift Pickup Complete Photos, EasyLift Futile Pickup Notification</i>) alongside PlastaGo's. So multi-brand is live today, confirming M1.1. Still to ask: which customers, and the kg-vs-m ² history	06-\$6
Q12	Hook bins (5/10/15 m³) — how does that service actually run? Delivered and left? Exchanged? Who owns the bin?	A whole service line with a dedicated 34T hooklift truck, absent from the website <i>and</i> the feature list. Different asset lifecycle to bags	06-\$4.2
Q13	F35 "recycling compliance" — what is it?	✅ EFFECTIVELY MOOT. It depends on EPA-licensed processing they don't do. Matt's own call: v3	09-\$20
Q14	NEW Which customers require the Site Risk	⚠️ An external dependency. We upload a PDF into <i>someone else's</i> system. Need the target portals, whether	M4.8

#	Question	Why it matters	Ref
	Assessment + SWMS? Which builder portals do the QR codes point at?	they need auth, and what happens if the upload fails	
Q15	NEW Xero configuration: which account codes, tax rates, and do you use tracking categories? Who invites our developers with admin rights?	Blocks M7.8. Matt confirmed on the call he'd arrange access	M7.8
Q16	"PrePaid Customer" — what is this account construct?	\$0 base with only exception charges invoiced. Needs its own handling, and it's the likely home for one-off/residential jobs	M5 · A.5
Q17	What recorded-vs-tip-off variance is acceptable before someone investigates?	Sets the tolerance threshold on the M4.4 review queue	M4.4
Q18	Is tip-off per run, per day, or per load? Do drivers ever tip mid-run?	The deduct-and-average algorithm assumes per run . Mid-run tipping breaks it	M4.4
Q19	Does anything else go in the truck? Other materials, other brands, mixed loads?	Mixed loads would invalidate the apportionment entirely	M4.4
Q20	Who else has TransVirtual logins besides you and the two drivers?	User migration and the pre-cutover invitation list	M10, M5 · C.1
Q34	NEW Which CRM do you use for proposals, and who processes your material?	Two loose ends: a third subscription we didn't know about (Matt: " <i>a separate CRM that's f***ing terrible</i> ") with possible lead data to migrate; and the identity of the third-party processor, which affects Diversion Certificate wording	Ø9- \$27, \$20
Q35	NEW Can you send a sample Diversion Certificate?	Decoded asked on Call 2 and it hasn't arrived. Blocks M9.5 — we need the exact data points and wording	M9.5
Q36	NEW Which customers require the Site Risk Assessment?	Matt: it applies " <i>on certain sites</i> ", not universally. Determines whether the SRA is a per-account or per-site flag, and how many builder QR portals we face	M4.8, Q14

12.4 ● TIER 3 — Useful, can be answered during the build

#	Question	Why we're asking
Q21	Invoices 104078 / 104079 — identical iPlasta jobs, same day, same 974 m ² , same \$90 contamination.	Migration dedupe logic

#	Question	Why we're asking
	Duplicates, or two units on one site?	
Q22	Contamination — binary flag, or do you want type and estimated %?	Today it's a yes/no. Grading makes it reportable and identifies repeat-offender sites
Q23	Do you need recurring / scheduled pickups? A large site may need one every fortnight for a year	Today that's 26 separate bookings. Cheap to add if wanted
Q24	Does board type matter (standard / fire-rated / wet-area / foil-backed)?	Foil-backed board is a known contaminant in gypsum recycling. You already capture brand (CSR/Knauf/Siniat)
Q25	Should we capture a site contact phone number? The current form doesn't have one	The driver is 5 minutes away and needs a gate opened — email is useless for that
Q26	Required-photo prompts — configured per site, per customer, or globally?	M4.5
Q27	Do builders ask you for diversion certificates today? How do you produce them?	Validates M9.5, the highest-value differentiator
Q28	Any plans outside NSW? The booking form has a State field	Zone and pricing configurability
Q29	BrickGo — what is it, and on what timeline?	Confirms the multi-material dimension
Q30	Subcontractor per-job pay rates — where are they held today?	Prep for RCTI in v1.1; the data model should anticipate it
Q31	Simvoly and Cognito Forms — notice periods / renewal dates?	Decommissioning Cognito after cutover
Q32	Who buys the recovered gypsum, and how are those sales managed?	Entirely invisible to software. Scopes the v1.1/v2 outbound product
Q33	What EPA licences do you operate under?	✅ ANSWERED: none. Transfer station, unlicensed, processing outsourced. F35 → v3
Q37	 What registration periods are in use — 12 / 6 / 3 months?	F43 rego reminder roll-forward (v1.1)

12.5  **ANSWERED** — closed, kept so they aren't re-asked

Question	Answer	Source
Xero or MYOB?	Xero. "I'm not looking to replace Xero, I'm looking to interface with it." MYOB presumed dead	Call
How is per-job weight measured?	Crane scale on bagged jobs (~60–70%). Hand-load jobs (~30%) cannot be weighed and are imputed from the tip-off remainder	Call

Question	Answer	Source
Is weight billed?	No. Priced on m ² ; weight exists to make the m ² →kg estimate better, and to audit the monthly tip bill	Call
Are drivers subcontractors? Is RCTI live?	All drivers are subcontractors. RCTI is planned, not live — weekly, per-job, exports to Xero as a bill	Call
Stripe / card payments?	Dropped by Matt. <i>"a lot of our payments come through direct deposit... happy to take that requirement out"</i>	Call
Email provider?	Microsoft 365 SMTP , not SendGrid. (Caveat: throttling + weaker bounce handling — raised)	Call
Native app or PWA?	PWA for customers — <i>"I don't think anyone's going to install a PlastaGo app."</i> Driver app PWA unless Q6 forces native	Call
Rebuild the website?	 RESOLVED — NO. The transcript predates the emails; the 14 Aug email governs : keep the Simvolvy site for SEO and bolt the app on. Scope Call 2 stands	Email (later)
What is the rate card?	Sydney $\$220 + \$0.16/\text{m}^2$ · Wollongong $\$250 + \$0.18/\text{m}^2$ · Newcastle $\$250 + \$0.20/\text{m}^2$ · bags $\$30$. 8 cards, tiers 1–4, effective-dated. Validated to the cent against 6 real invoices	Email + tenant
Run sheets vs route optimisation?	Different things. Run sheets = <i>"going to be a necessity"</i> . Route optimisation = <i>"very nice to have"</i> (and is really succession planning)	Call
Bag tracking priority?	Version 3 , Matt's own call	Call
Chat scope?	Office ↔ driver only. No driver-to-driver, no group channel	Call
Where does job duration come from?	Arrived → Complete Job , driver-tapped	Call
Customer feedback / ratings?	Nice-to-have	Call
Environmental impact analytics?	v2. <i>"This is not important... one of those AI suggestions"</i>	Call
Contamination dispute process?	Driver marks contaminated → Additional Service Approvals queue → office approves → \$90	Tenant
Where is the facility? EPA licence?	 A yard run as a transfer station. NO EPA licence. Processing is outsourced (hence the monthly tip bill). On-site processing is a future plan	Call 2
What is the real SLA?	 5 business days from the customer's ready date — not "3–5 days from request". Most jobs have no time; a minority have a hard permit window	Call 2

Question	Answer	Source
What does the Diversion Certificate contain?	✓ Matt's name for it. Per job , showing both m² and tonnage diverted from landfill. For Green Building Council / NABERS reporting	Call 2
Invoice formats?	✓ PDF only . CSV was for one customer they no longer service	Call 2
What does F43 need?	✓ Odometer + expense log → cost per km ; rego reminder 2 weeks out . No parts/labour breakdown	Call 2
What does F53 need?	✓ Licence / white card / crane tickets + expiry, 1-month reminder, admin-managed licence types	Call 2
What is F58 really?	✓ Proximity dispatch — flag a late booking to a driver already in that area. Not availability scheduling	Call 2
Does he want invoice-template priority + filters?	✓ No . "I don't need this level of granularity." Simplify to per-customer assignment. But he does want layout editing — see Scope Call 1	Call 2
Volumes, accounts, drivers, fleet?	5,079 jobs · ~135–160/mo · 53 accounts (31 active in last 12 mths, top 5 = 87%) · 2 drivers · 5 vehicles	Tenant

13. The 20-day plan — design lock, then two backends

Confirmed approach: days 1–3 are frontend only. No backend, no mobile development. The **entire web UI** is built against mock data and shown to Matt on **day 3**. Once the design is confirmed, the same design is applied in Flutter and the backend is built for both frontends.

13.1 Why this sequencing is right — and what it costs

What it buys:

- **The design is locked before any backend or Flutter investment**, so rework is cheap
- **The API contract falls out of real UI needs** rather than being guessed (\$6A.9)
- **Flutter mirrors a confirmed design** instead of a moving target — critical, because the mobile app is a separate repo and a separate language
- Matt sees something real at the 15% mark, which is the right moment for a deposit-funded engagement

What it costs — state this plainly:

⚠ **The pricing regression harness moves from days 3–4 to days 6–8.** It is Risk 1, the highest-rated risk in the project, and it now starts later. **That is acceptable but must be watched: it cannot slip past day 10 (halfway).** With MongoDB carrying no database-level integrity guarantees (\$6A.3), this harness is the only thing standing between us and a mis-priced invoice.

13.2 Days 1–3 — frontend only

Who	Work
Web dev 1 + Web dev 2	Day 1: design system, Tailwind/shadcn setup, PlastaGo brand applied (logo and palette are in the client's asset pack), screen inventory, Zod schemas in packages/shared . Days 2–3: every screen, against MSW mock data
Mobile dev	⚠️ Apple Developer enrolment — START DAY 1 (Risk 8, longest external lead time). Flutter project scaffold, CI, device setup. Write the offline sync protocol spec (§6A.4) — this is design-independent, so it is the right work for these three days

Screens to build — roughly 43:

Surface	Screens
Admin console (~22)	Job list · job detail · create job · accounts · account detail · sites · allocation board · run sheets · futile review queue · additional-service approvals · awaiting-PO queue · PO review queue · invoice list · invoice detail · template assignment · leads/onboarding queue · vehicles · drivers/licences · reports (×4) · dashboard · settings
Customer portal (~11)	Dashboard · book a pickup · edit pickup · job history · job detail with photos · invoices · monthly reports · Diversion Certificates · site supervisors · account preferences · sites
Driver PWA (~10)	Login (OTP) · today's run sheet · job detail · status actions · weights + tip-off · photo capture · futile · contamination · Site Risk Assessment form · pre-start checklist
Auth	OTP request · OTP verify · role-based landing

Judgement call — the driver screens ARE in the Day-3 demo. They are web (a Vite PWA), and more importantly **Flutter will mirror them**. Starting Flutter against an unconfirmed design defeats the whole purpose of the design lock. It does add screens to a tight three days — if something has to give, cut the four report screens, not the driver app.

Deliverables at end of day 3:

1. ☒ Complete, navigable web UI on mock data
2. ☒ **API contract** — Zod schemas → generated OpenAPI, **frozen** (§6A.9)
3. ☒ **Offline sync protocol spec** — one document, two future implementations
4. ☒ Apple enrolment in flight

What "complete frontend in 3 days" means, precisely: screens, navigation, layout, component library and mock data. **It does not yet mean** wired state, validation rules, error and empty states, optimistic updates or the offline layer — those are the expensive parts and they come from day 4. Say this in the demo (§13.5).

13.3 ★ Day 3 — the show-and-tell

Show, in this order — familiar first, new last:

1. **Job list and job detail** — his #1 and #2 daily screens. Parity, instantly.
2. **Create job** — with the **live price estimate** TransVirtual cannot give him.

3. **Allocation board and run sheet** — visual, and immediately legible to an ex-driver allocator.
4. **The driver PWA on an actual phone.** Hand it to him. This is the moment it becomes real — and it is the design Flutter will mirror, so **this is the screen set that most needs his sign-off.**
5. **The customer portal** — the thing TransVirtual has never given him.
6. **A sample invoice** with $2 \times \$30 = \60 line items, and a **Diversion Certificate.**

Use the session to close open Tier 1 questions — a screen in front of him gets answers an email will not:

Show him	Closes
Sample invoice + the template options	Q5 — the authoring decision (Scope Call 1)
The booking flow	Q7 — who actually raises bookings
A draft Diversion Certificate	Q35 — we still do not have his sample
The driver screens	Design sign-off that unblocks all Flutter work
The migration plan on screen	Q1 — history vs archive

13.4 Days 4–20 — three tracks

Days	A — Backend (web dev 1)	B — Frontend wiring (web dev 2)	C — Flutter (mobile dev)
4–5	Mongo schemas + Mongoose models · \$jsonSchema validators · repository layer · OTP auth + JWT	Wire admin console to the real API · validation · error/empty states	Flutter shell · navigation · auth · local DB (Drift)
6–8	⚠️ PRICING ENGINE + REGRESSION HARNESS — 300 real invoices, 100% match, in CI	Accounts · sites · job CRUD · allocation board	Run sheet · status flow · Arrived → Complete clock
8–10	Additional services · tip-off deduct-and-average · exception-queue logic	Exception queue UIs — futile, approvals, awaiting-PO	Weights · tip-off entry · photos + required-photo prompts
10–12	Invoicing · two-invoice flow · template assignment · PDF generation	Invoicing UI · customer portal wiring	Futile · contamination · SRA → PDF + SWMS merge → QR → builder portal
12–14	Xero push/pull · ★ AI PO pipeline (M365 mailbox + Mistral OCR)	Driver PWA offline layer — IndexedDB/Dexie, service worker, sync	Background geolocation · push · native build
14–16	PO pipeline tuning against real PO emails · notifications · Diversion Certificates	M5 Part 2 enrolment · dashboards · reports	iOS distribution (TestFlight/Ad Hoc) · Android APK · device testing
16–18	MIGRATION — accounts, jobs, invoices, photos · parallel-run validation	Migration QA · reconciliation vs TransVirtual reports · F43 / F53 / F22 screens	Field test — both drivers work a real run on both builds

Days	A — Backend (web dev 1)	B — Frontend wiring (web dev 2)	C — Flutter (mobile dev)
18–19	Invitation campaign (T-5 → T-1) · top-5 account walkthroughs	UAT with Matt + office	UAT fixes with both drivers
19–20	Fixes · cutover runbook · GO LIVE		

Go-live gates — all three must be green:

1. **Pricing harness passing on 300 real invoices**
2. **Top five accounts have activated users** (M5 · C.2)
3. **Both drivers have completed a real run** on the app they will actually use

13.5 Sequencing rules that must not be broken

1. **Apple Developer enrolment starts Day 1.** Longest external lead time, zero cost to start early (Risk 8).
2. **The API contract freezes at day 3.** Two consumers depend on it; changes after that need a version bump (§6A.9).
3. **The offline sync protocol is written before either implementation starts** — days 1–3, by the mobile dev (§6A.4).
4. **The pricing harness lands days 6–8 and cannot slip past day 10.** With MongoDB providing no integrity backstop, this is the only guard on invoice correctness (§6A.3).
5. **Real PO emails needed by day 12** — ask Matt in week 1 (Risk 9).
6. **Migration gets two full days (16–18), not an afternoon.** The extra six days over the original plan exist partly to buy exactly this.

13.6 How to frame the Day-3 demo — this matters

A complete UI at day 3 will look far more finished than the system is. Matt is an operations director; he will see working screens and reasonably conclude the build is most of the way there. **The risk on this project was never the interface** — it is the pricing engine, the AI extraction, the migration and the cutover, none of which are visible.

Open the session with this:

"What you're seeing today is the complete look and feel, and it's real code — but it's running on sample data, with no engine behind it yet. That's deliberate: we wanted your sign-off on every screen before we build the backend twice, once for the web and once for the drivers' phones. The next seventeen days are the parts you can't see — making sure every invoice matches what TransVirtual would have produced, teaching the system to read your POs reliably, getting three years of jobs and photos across, and switching over without breaking your billing. That's where the risk is, and that's where the time goes."

That paragraph prevents the *"but it looked finished two weeks ago"* conversation on day 17.

14. How to present this to the client

Lead with what they get. Then draw the boundary honestly.

"In 20 days you'll be live on a system that replaces the six screens you use in TransVirtual every day — creating jobs, finding jobs, approving driver charges, working futile pickups, invoicing, and seeing where your drivers are.

On top of that you get two things TransVirtual has never been able to give you: a customer portal your five biggest accounts can log into and book, track and self-serve from — and automated recycling certificates they can use for Green Star.

Your pricing will be validated against your last 300 real invoices before we switch anything over, and we'll keep TransVirtual read-only for a fortnight as a safety net.

Route optimisation, the PO-reading AI, RCTI and the drag-and-drop invoice designer are all real and we want to build them — they're v1.1, right after go-live. Putting them into the first 20 days would put your invoicing at risk, and that's the one thing we won't do."

Anchor the commercial case on:

- **~\$9k/year** of contamination charges sitting unapproved in a queue right now
- **A futile pickup (\$120) unactioned since August 2025**
- **Retention of five accounts worth 87% of revenue** — the portal is aimed at them
- **Cash cycle** — POs enforced, invoices out same-day, payment status visible

Never anchor it on replacing TransVirtual's subscription. The value is recovered revenue, recovered owner-hours and customer retention — not saved licence fees.

15. Cross-references

- [09-CLIENT-CALL-FINDINGS.md](#) — ★ **highest-authority source.** The discovery call: tip-off algorithm, Site Risk Assessment workflow, RCTI mechanics, confirmed/dropped integrations
- [06-TRANSVIRTUAL-INCUMBENT-SYSTEM.md](#) — the incumbent, live volumes, configuration, migration inventory
- [07-PRICING-AND-BILLING-MODEL.md](#) — the rate engine in full, real invoice data, the template trap
- [03-BOOKING-FORM-TEARDOWN.md](#) — intake requirements, validated against production data
- [04-INDUSTRY-REGULATORY-COMPETITIVE.md](#) — EPA obligations behind F35 and the certificates
- [05-OPERATIONAL-WORKFLOW-AND-PAIN-POINTS.md](#) — value leakage map